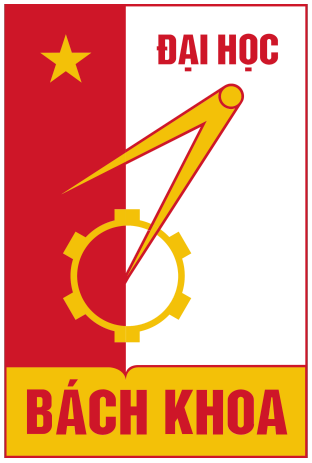
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY



A Comparative Study of Statistical, Machine Learning, and Deep Learning Models for Multivariate Weather Forecasting

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Project Report

Course: IT2022E - Applied Statistics and Experimental Design

Group Members:

No.	Full Name	Student ID
1	Nguyen Le Anh	202416663
2	Nguyen Minh Tuan	202416759
3	Dang Hoang Viet	202416765
4	Pham Quang Vu	202416767
5	Bui Nguyen Vu	202416768

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Chapter 1

Introduction

1.1 Problem Motivation

Weather strongly influences agriculture, energy demand, transportation, public safety, and everyday decision making. Accurate short-term forecasts of variables such as temperature, humidity, and wind enable better planning and risk mitigation. While operational numerical weather prediction relies on solving the governing physical equations on supercomputers, data-driven models trained on historical observations offer a lightweight, station-level complement that can be deployed cheaply and retrained frequently. This project studies how well a range of statistical, machine-learning, and deep-learning models can forecast the next 24 hours of local weather from recent history.

1.2 Importance of Weather Forecasting

Reliable forecasting supports energy load balancing, renewable-generation scheduling, agricultural irrigation planning, and early warning for adverse conditions. Even modest reductions in forecast error translate into measurable operational and economic benefits.

1.3 Research Objectives

The objectives of this project are to:

1. Establish a single, shared experimental protocol (data handling, split, horizon, and metrics) so that heterogeneous models are compared fairly.
2. Implement and evaluate five forecasting models: DLinear, SARIMA, VAR, XGBoost, and LSTM.
3. Provide a detailed treatment of the XGBoost model, including feature engineering tailored to multivariate hourly time series.

4. Quantify and analyse forecast accuracy overall, per forecast horizon, and per target variable.

1.4 Project Scope

The study targets *short-term* forecasting: a fixed 24-hour horizon from a fixed look-back window, for a single location described by hourly observations. The seven forecast targets and the evaluation procedure are held constant across all models. Long-range (sub-seasonal/seasonal) forecasting, spatial fields, and ensemble physical models are out of scope.

1.5 Report Organisation

Chapter 2 describes the dataset. Chapter 3 presents an exploratory-data-analysis template. Chapter 4 details preprocessing. Chapter 5 presents the five forecasting models, with XGBoost covered in depth. Chapter 6 defines the evaluation protocol and metrics, then reports and discusses the results.

Chapter 2

Dataset Description

2.1 Data Source

The dataset is an hourly weather export in the Open-Meteo style [8], stored as `weather_dataset.csv`.

2.2 Data Collection Period

The data span **2021-01-01** to **2025-12-31** at hourly resolution, giving **43,824** rows with no gaps and no missing values.

2.3 Features Description

Of the raw hourly export, this study uses the columns in Table 2.1: five variables that are modelled directly plus wind speed and direction, which are combined into the Cartesian wind components `wx`/`wy` (Chapter 4). The raw file also contains precipitation, layered (low/mid/high) cloud cover, wind gusts, and topsoil moisture; the group agreed to exclude these from the seven-variable forecasting target set, so they are not used in any model and are not analysed further.

Table 2.1: Dataset columns used in this study and their clean names.

Raw column	**Clean name**	**Description**
`temperature_2m (°C)`	`temp`	Air temperature at 2 m (target)
`relative_humidity_2m (%)`	`rh`	Relative humidity at 2 m (target)
`dew_point_2m (°C)`	`dew`	Dew point at 2 m (target)
`surface_pressure (hPa)`	`pressure`	Surface pressure (target)
`cloud_cover (%)`	`cloud`	Total cloud cover (target)
`wind_speed_10m (km/h)`	`wind_speed`	Wind speed at 10 m (input for `wx`/`wy`)
`wind_direction_10m (°)`	`wind_dir`	Wind direction at 10 m (input for `wx`/`wy`)
`wx`, `wy` (km/h)		Cartesian wind components, engineered (targets)

2.4 Target Variable Description

All models forecast the same seven targets:

`temp`, `rh`, `dew`, `pressure`, `cloud`, `wx`, `wy`,

where `wx` and `wy` are the Cartesian wind components derived from wind speed and direction (Chapter 4). Each forecast is a 24×7 block (24 hours ahead $\times$ 7 variables).

2.5 Dataset Characteristics

The series is strictly hourly and complete, which simplifies windowing and avoids imputation in the common case.

Chapter 3

Exploratory Data Analysis (EDA)

This chapter analyses the cleaned dataset to characterise its distributions, dependencies, and temporal structure, and to motivate the preprocessing and modelling choices in later chapters. All figures are produced by `report/eda/run_eda.py`.

3.1 Dataset Overview

The dataset is complete and strictly hourly (Table 3.1): five full calendar years with no missing timestamps and no missing values.

Table 3.1: Dataset overview.

Property	**Value**
Number of samples (rows)	43,824
Number of columns used	7 (5 direct + wind speed/direction)
Number of target variables	7 (`temp,rh,dew,pressure,cloud,wx,wy`)
Temporal resolution	Hourly
Data period	2021-01-01 – 2025-12-31
Missing values	0

3.2 Missing Value Analysis

A column-wise scan finds **zero** missing values across all variables and all 43,824 hourly timestamps; the hourly index is also gap-free after resampling. Imputation is therefore not required, and the time-interpolation step in Chapter 4 acts only as a safeguard. A missing-value heatmap is omitted as it would be uniformly empty.

3.3 Statistical Summary

Table 3.2 reports descriptive statistics for the seven targets. The profile is that of a warm, humid, monsoonal climate: a high mean temperature (24.0 °C), very high relative humidity (mean 78.8%, median 81%), persistently high cloud cover (mean 73.4%, median 99%), and surface pressure tightly concentrated around 1010 hPa (std 7.4). The wind components are centred near zero (`wx` mean 0.3, `wy` mean 3.8 km/h) with a wide spread.

Table 3.2: Summary statistics for the seven targets (full 2021–2025 period, native units).

Variable	Mean	Median	Std	Min	Max
`temp` (°C)	24.01	24.90	5.56	6.30	39.00
`rh` (%)	78.75	81.00	14.86	22.00	100.00
`dew` (°C)	19.81	22.10	6.36	-8.90	28.80
`pressure` (hPa)	1009.67	1009.30	7.41	982.00	1032.90
`cloud` (%)	73.41	99.00	37.26	0.00	100.00
`wx` (km/h)	0.28	0.00	8.16	-22.69	57.23
`wy` (km/h)	3.83	4.25	4.66	-36.69	45.02

3.4 Distribution Analysis

The marginal distributions (Fig. 3.1) and boxplots (Fig. 3.2) confirm several features that shape modelling: `cloud` is strongly bimodal, piling up at 0% and 100%; `rh`, `temp`, and `dew` are left-skewed toward their upper range; `pressure` is approximately symmetric; and the wind components `wx`, `wy` are roughly symmetric about zero with heavy tails. The boxplots show that the heavy "outliers" are concentrated in the wind components — genuine meteorological extremes (gusts/storms) rather than sensor errors — which supports keeping them and using an outlier-robust loss (MAE).

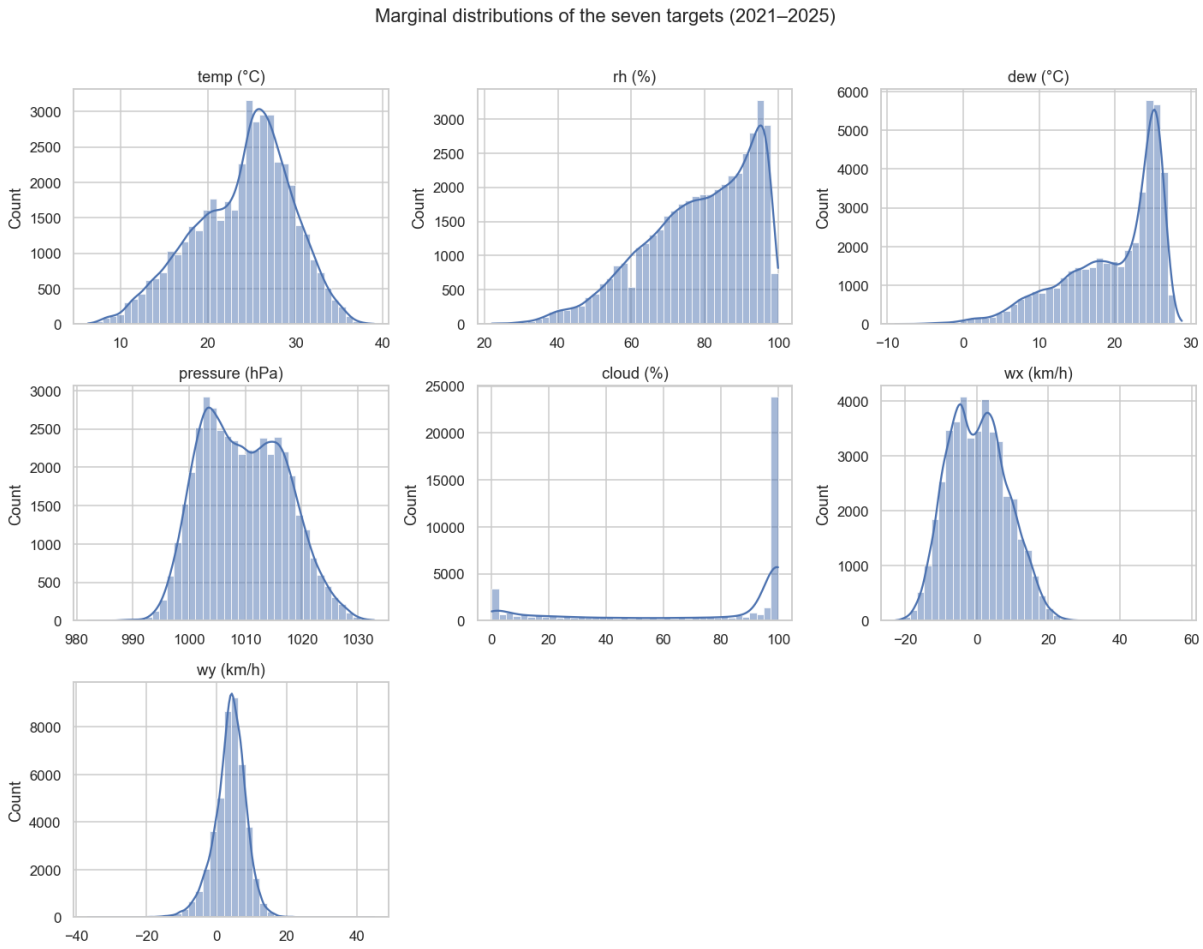


Figure 3.1: Marginal distributions of the seven targets.

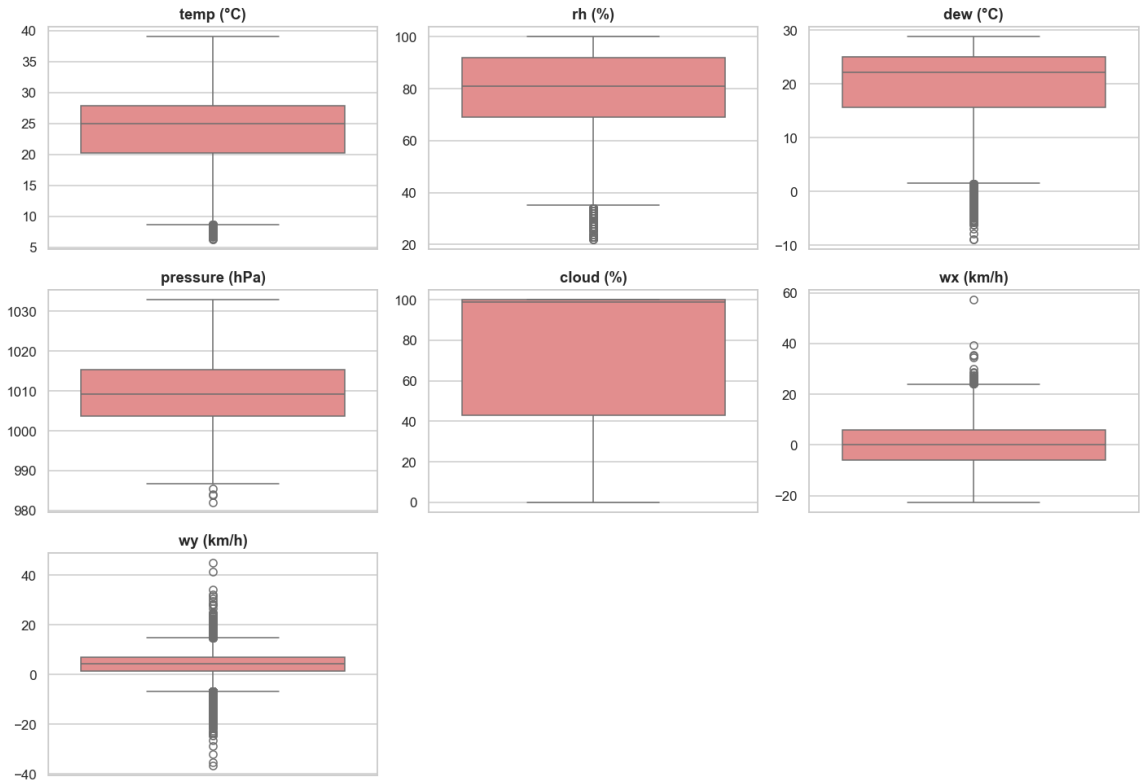


Figure 3.2: Per-variable boxplots of the seven targets for outlier inspection.

3.5 Correlation Analysis

The correlation matrix (Fig. 3.3) reveals strong, physically sensible dependencies: `temp`–`dew` (+0.85), `dew`–`pressure` (−0.87), and `temp`–`pressure` (−0.81). Relative humidity correlates moderately with dew point (+0.52), while `cloud` and the wind components are only weakly linearly related to the other targets — an early indication that `cloud` and wind will be the hardest to forecast. These inter-variable links justify a *multivariate* approach and the lag/rolling features used by XGBoost.

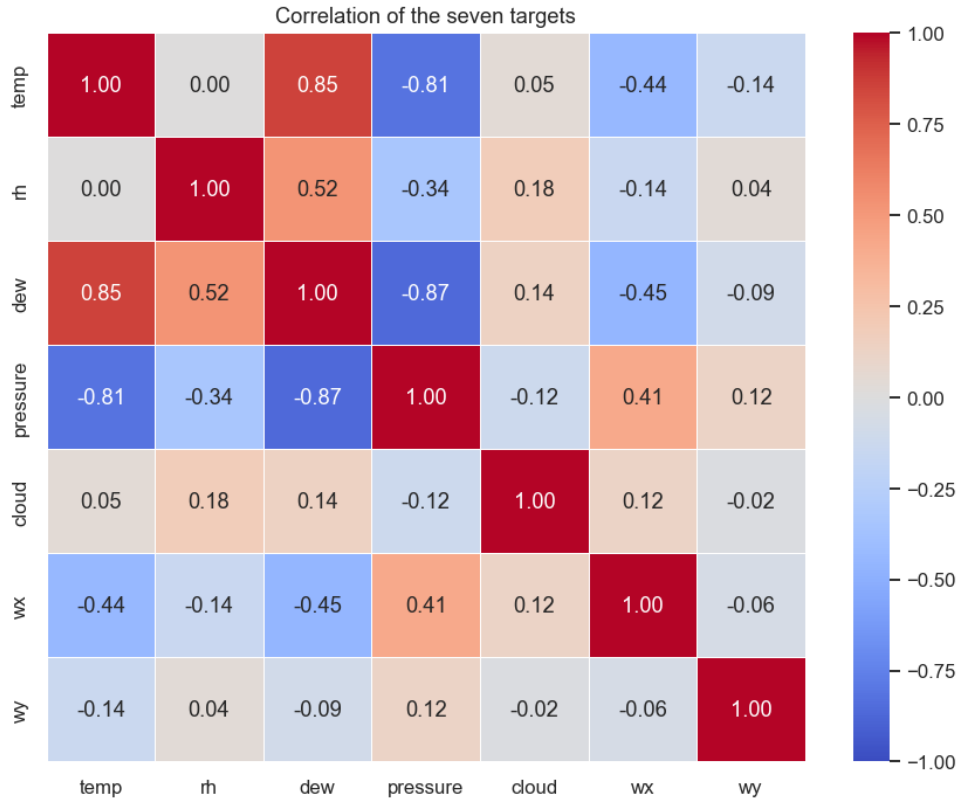


Figure 3.3: Correlation matrix of the seven targets.

3.6 Temporal Analysis

Weekly means (Fig. 3.4) show a clear annual cycle in temperature and humidity and a milder seasonal swing in pressure. Aggregating by month and by hour-of-day (Fig. 3.5) exposes two dominant periodicities: a seasonal cycle across the year and a strong diurnal cycle (afternoon maximum, pre-dawn minimum). These two cycles directly motivate the cyclical hour-of-day and day-of-year encodings in Chapter 4.

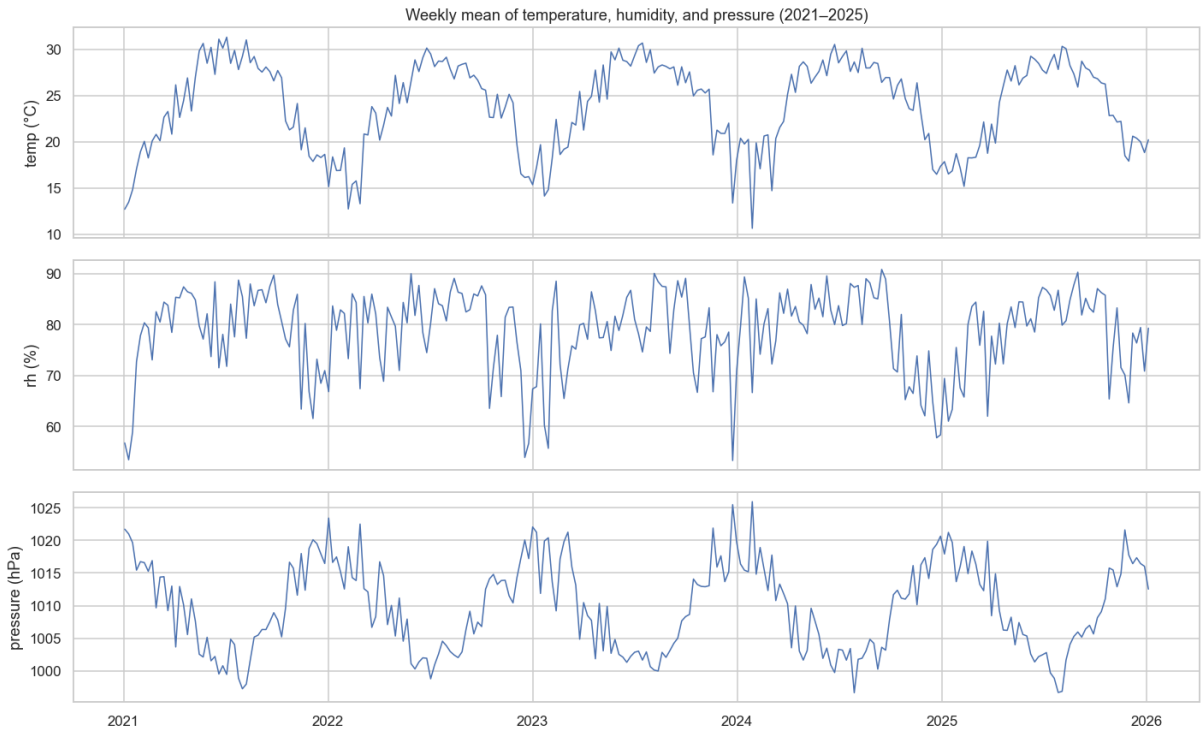


Figure 3.4: Weekly mean of temperature, humidity, and pressure over the full period.

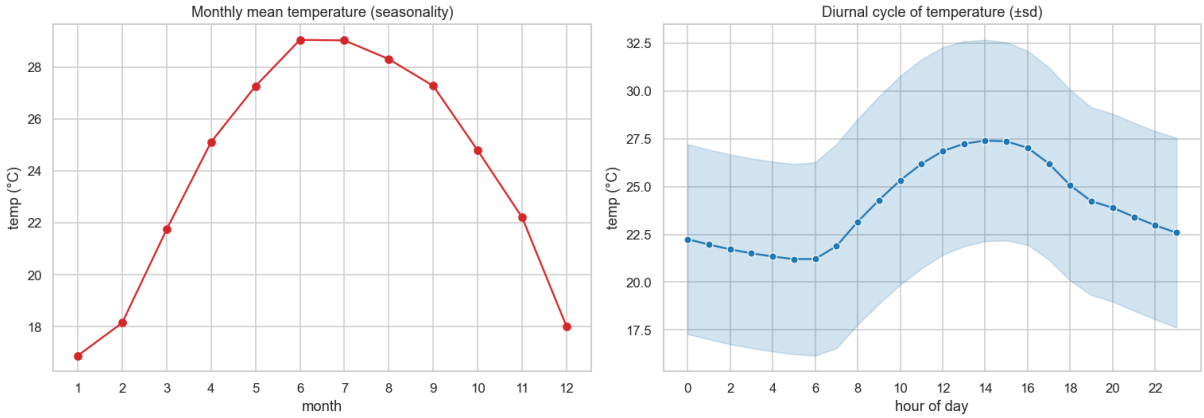


Figure 3.5: Monthly seasonality (left) and diurnal cycle (right) of temperature.

3.7 Target Variable Analysis

An additive seasonal decomposition of daily-mean temperature (Fig. 3.6, period = 365 days) separates a dominant annual seasonal component (amplitude $\approx \pm 5\,^{\circ}\mathrm{C}$) from a slow trend and a small residual. The large, regular seasonal term confirms that much of the variance in the targets is predictable structure, which all models should capture; the residual term bounds the irreducible short-term noise.

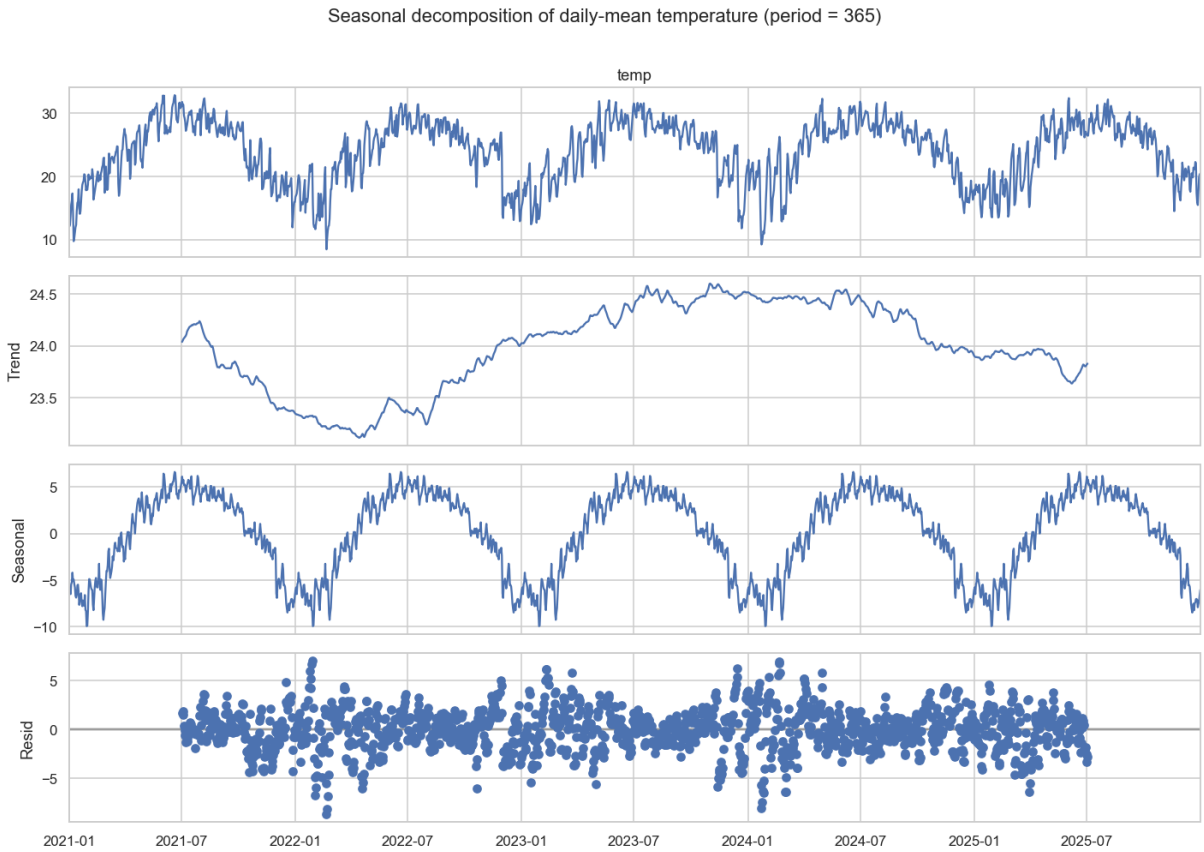


Figure 3.6: Seasonal decomposition of daily-mean temperature.

Extending this to all seven targets (Fig. 3.7) shows that each variable carries a clear annual seasonal component: humidity and cloud cover have the largest annual swings, the wind components the smallest, and every trend term is mild and non-monotonic. The bulk of each series is therefore predictable seasonal structure on top of short-term noise.

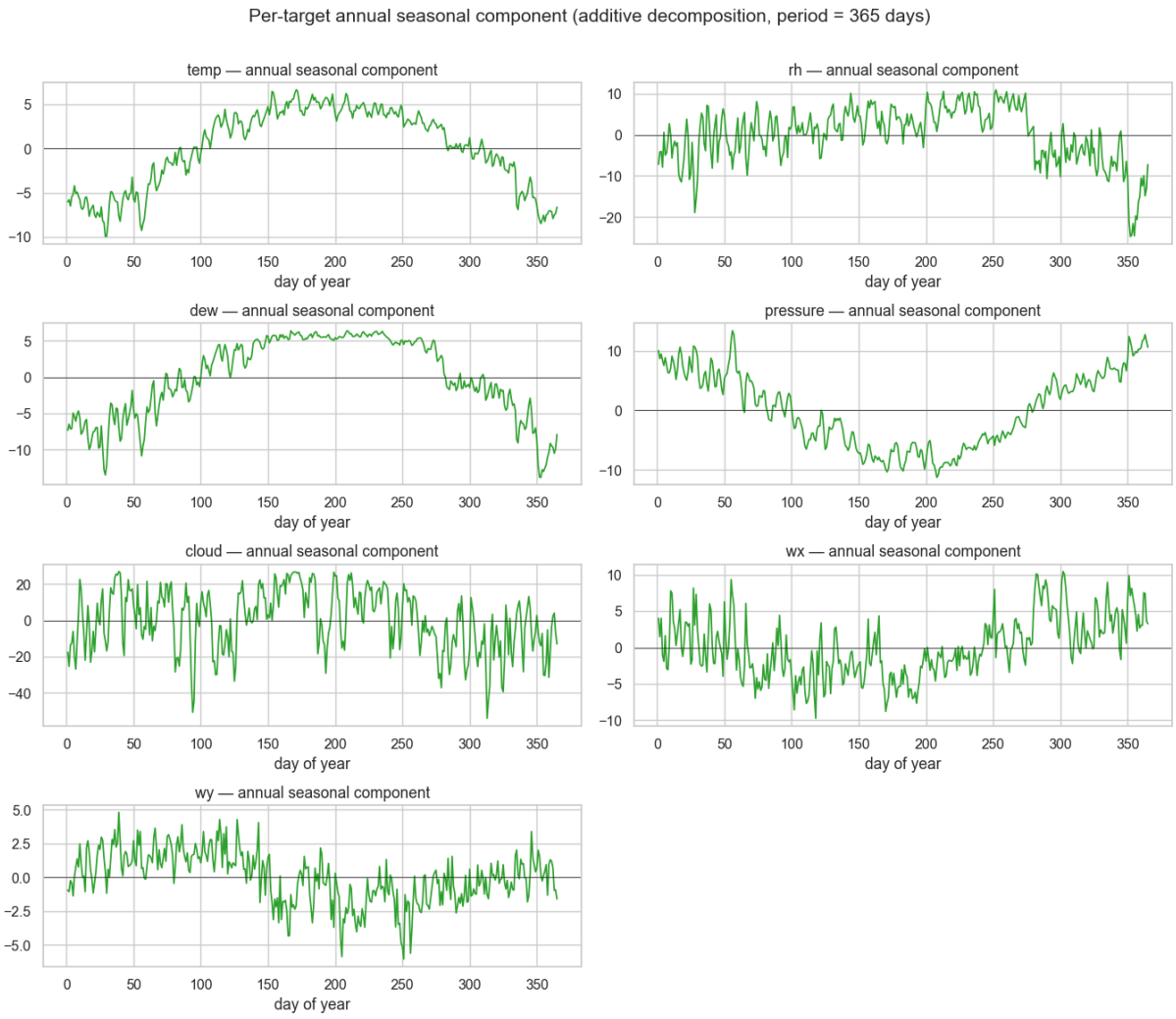


Figure 3.7: Annual seasonal component of each target (additive decomposition, period = 365 days).

3.8 Time-Series Structure

This section adds the time-series diagnostics that directly inform the statistical models (especially VAR, Section 5.3): stationarity, autocorrelation structure, the diurnal cycle, and lagged cross-dependence.

3.8.1 Stationarity

Two complementary tests are applied to each target: the Augmented Dickey–Fuller (ADF) test, whose null hypothesis is a unit root (small $p \Rightarrow$ stationary), and the KPSS test, whose null hypothesis is stationarity (large $p \Rightarrow$ stationary). Using both guards against the weaknesses of either alone. The results (Table 3.3) show that ADF rejects the unit root for all seven targets at $p \leq 10^{-12}$, and KPSS agrees (fails to reject stationarity) for five of them. For `temp` and `wx`, KPSS marginally rejects ($p = 0.049$ and 0.030); combined with

the decisive ADF rejection, this points to *deterministic* annual seasonality rather than a stochastic trend — a pattern also visible as the wavy-but-flat rolling mean in Fig. 3.8, whose rolling standard deviation stays roughly constant. Differencing is therefore not required, justifying the $d = 0$ choice used by the VAR model.

Table 3.3: Stationarity tests on the seven targets (full 2021–2025 series).

Variable	**ADF** p	**KPSS** p	**Verdict**
`temp`	6.6×10^{-13}	0.049	stationary (borderline KPSS: annual cycle)
`rh`	4.3×10^{-29}	≥ 0.10	stationary
`dew`	2.1×10^{-14}	≥ 0.10	stationary
`pressure`	6.5×10^{-13}	≥ 0.10	stationary
`cloud`	$< 10^{-30}$	≥ 0.10	stationary
`wx`	2.9×10^{-30}	0.030	stationary (borderline KPSS: annual cycle)
`wy`	$< 10^{-30}$	≥ 0.10	stationary

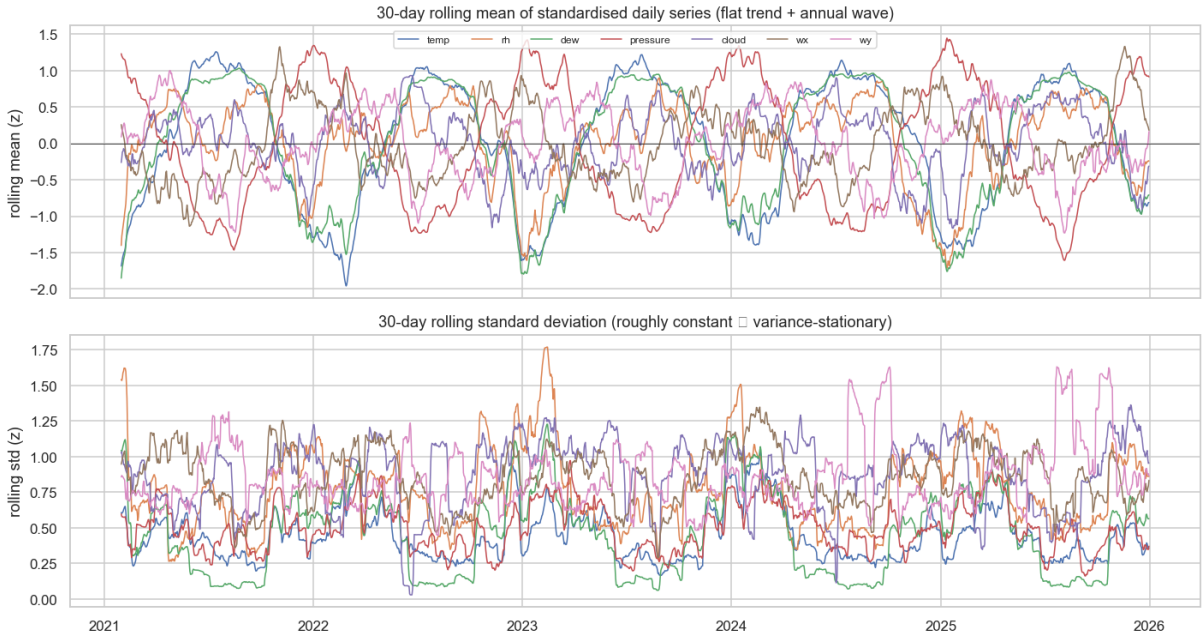


Figure 3.8: Rolling 30-day mean and standard deviation of the standardised daily series. The mean is flat apart from an annual wave (seasonality, not a unit-root trend) and the standard deviation is roughly constant.

3.8.2 Autocorrelation Structure

The autocorrelation (ACF) and partial-autocorrelation (PACF) functions (Fig. 3.9) characterise each target's memory. The ACFs decay slowly with a clear periodic bump near lag 24 (the daily cycle), while the PACFs cut off after a few lags, with an additional spike near lag 24. This "PACF cuts off, ACF tails off" signature indicates an autoregressive

structure of low base order plus daily-cycle lags, and cross-checks the AIC-selected lag order used by VAR (Section 5.3.3).

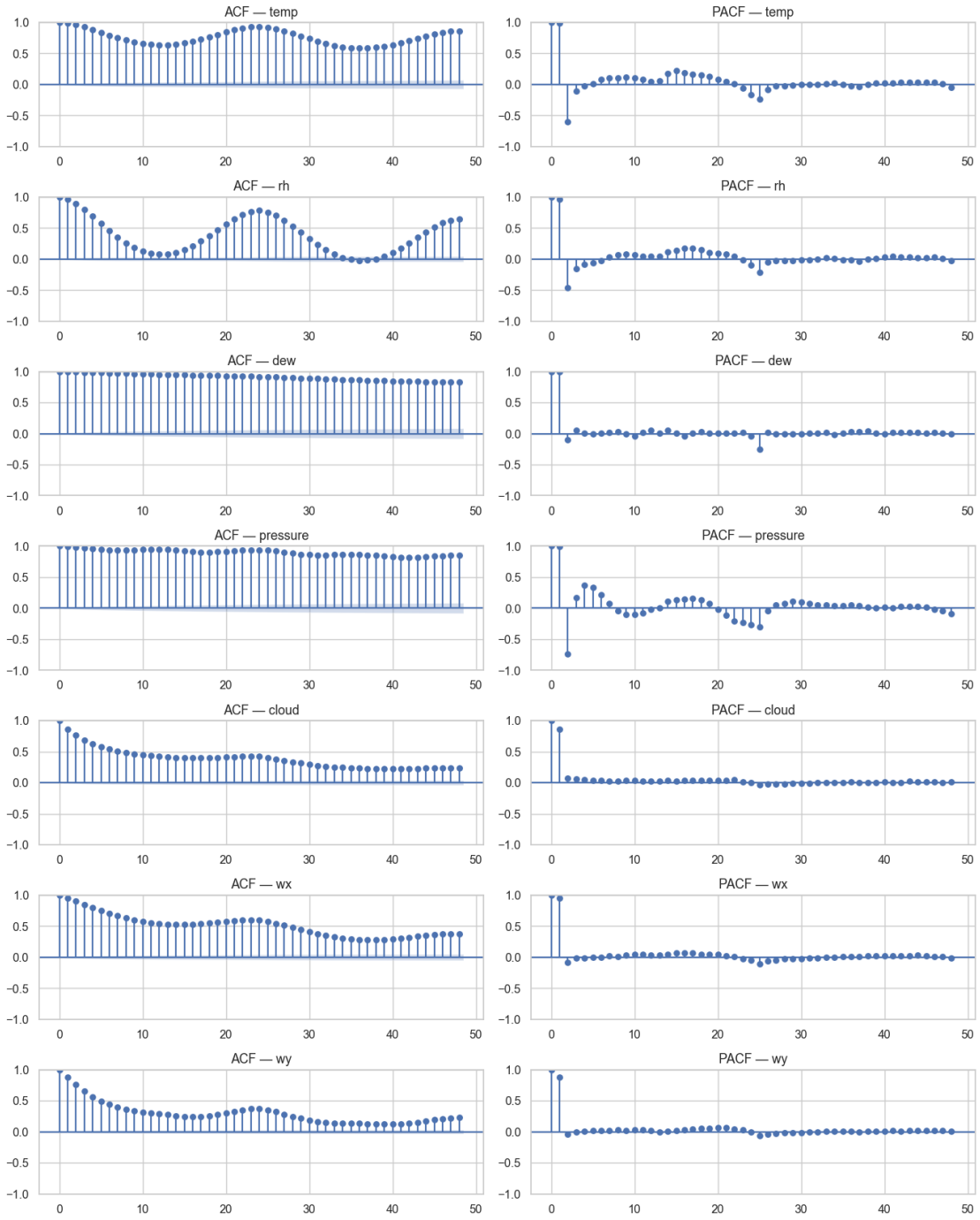


Figure 3.9: ACF (left) and PACF (right) of each target, lags 0–48 hours.

3.8.3 Diurnal Cycle

Standardising each target by hour of day (Fig. 3.10) exposes the 24-hour cycle directly: `temp` peaks in the afternoon and troughs before dawn, `rh` moves in near-perfect anti-phase, and the wind components show a clear daytime/nighttime pattern, whereas `cloud`

and `pressure` have weaker, phase-shifted cycles. The afternoon temperature peak near 07–08h on the x-axis indicates the timestamps are in UTC for this location.

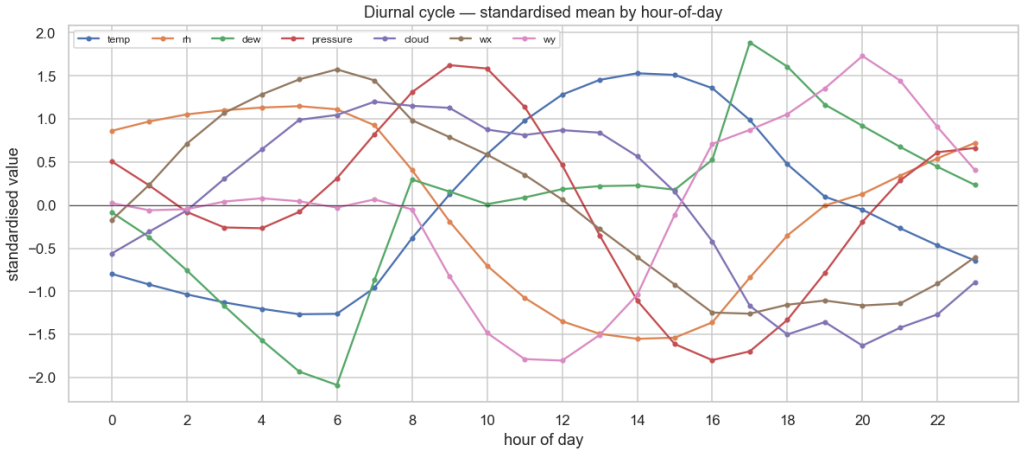


Figure 3.10: Diurnal cycle: standardised mean of each target by hour of day.

3.8.4 Lagged Cross-Correlation

Finally, the lagged cross-correlation matrix (Fig. 3.11) measures how each target's recent past ($t - 3\,\mathrm{h}$) relates to each other target's present. Strong off-diagonal entries appear among the thermodynamic variables — `dew`–`pressure` (-0.87), `temp`–`dew` ($+0.85$), and `temp`–`pressure` (-0.84) — meaning one variable's history carries predictive information about another. This lagged cross-dependence is exactly the signal a *multivariate* model (VAR) can exploit but seven independent univariate models cannot, and it motivates the multivariate approach taken throughout this study.

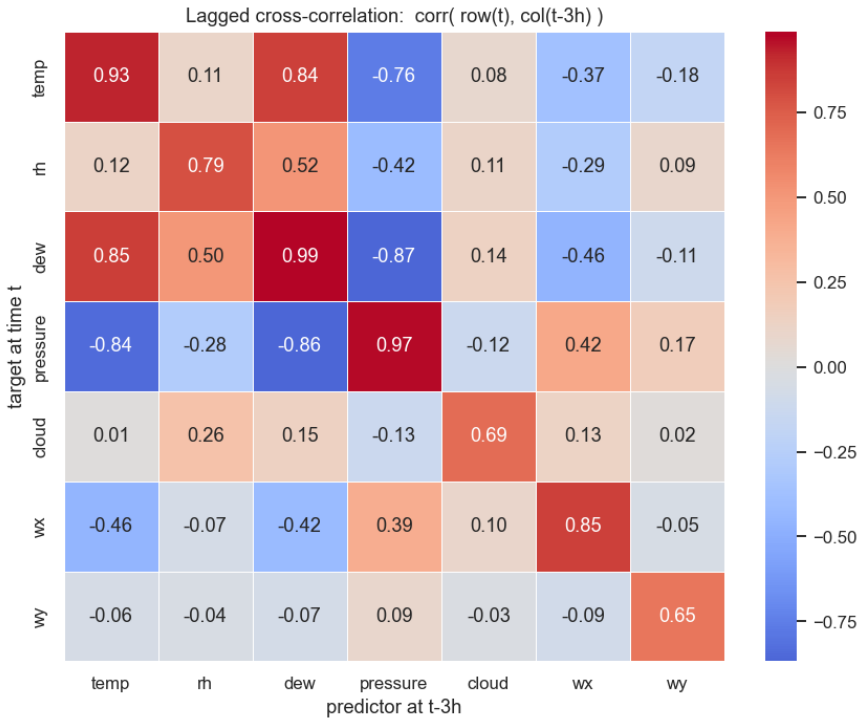


Figure 3.11: Lagged cross-correlation $\mathrm{corr}\big(\mathrm{row}(t),\, \mathrm{col}(t-3\mathrm{h})\big)$ among the seven targets.

3.9 Key Findings from EDA

- **Complete data** — no missing values or gaps, so no imputation is needed (Chapter 4 interpolation is a safeguard only).
- **Strong periodicity** — pronounced annual and diurnal cycles ⇒ cyclical time encodings are added as input features.
- **Strong inter-variable correlation** (temp/dew/pressure) ⇒ a multivariate model with lagged cross-features is appropriate.
- **Skewed/bounded targets** — `cloud` is bimodal and the wind components `wx`/`wy` take zero/near-zero values when calm ⇒ MAPE is unsafe for these, MAE is the primary metric, and `cloud` is expected to be hardest.
- **Genuine extremes** in the wind components ⇒ retained (not clipped, except the non-negativity guard on wind speed).
- **Stationary in levels** — ADF and KPSS agree the targets have no unit root (Table 3.3); the borderline KPSS cases reflect deterministic annual seasonality ⇒ no differencing ($d = 0$).

- **Low AR order + daily lag** — the PACF cuts off after a few lags with a lag-24 spike ⇒ a modest lag order augmented by daily-cycle terms suffices (cross-checks the AIC choice for VAR).
- **Lagged cross-dependence** — strong off-diagonal cross-correlations among `temp`/`dew`/`pressure` (Fig. 3.11) ⇒ a multivariate model can exploit structure unavailable to univariate ones.

Chapter 4

Data Preprocessing

The same preprocessing is applied to every model so the comparison is fair.

4.1 Data Cleaning

The `time` column is parsed to a timestamp, set as the index, sorted ascending, and resampled to a strict hourly frequency. As a sensor-error guard, wind speed is clipped to be non-negative,

$$\texttt{wind_speed} \leftarrow \max(\texttt{wind_speed}, 0),$$

which removes spurious sentinel values (e.g. -9999).

4.2 Missing Value Handling

Any incidental gaps are filled by time-based interpolation followed by forward- and back-filling:

$$\tilde{x}_t = \text{interp}(x)_t, \qquad \text{then } \texttt{ffill} \rightarrow \texttt{bfill}.$$

On the present dataset there are no gaps, so this step is a safeguard.

4.3 Wind Decomposition

Wind direction is circular and not directly learnable; it is converted to Cartesian components,

$$\texttt{wx} = \texttt{wind_speed} \cdot \cos\Big(\text{rad}(\texttt{wind_dir})\Big), \tag{4.1}$$

$$\texttt{wy} = \texttt{wind_speed} \cdot \sin\Big(\text{rad}(\texttt{wind_dir})\Big), \tag{4.2}$$

so that magnitude and direction are encoded smoothly.

4.4 Cyclical Time Encoding

Hour-of-day and day-of-year are encoded with sine/cosine pairs to respect their periodicity:

$$\text{hour}_{\sin} = \sin\left(\frac{2\pi\, h}{24}\right), \qquad \text{hour}_{\cos} = \cos\left(\frac{2\pi\, h}{24}\right), \tag{4.3}$$

$$\text{doy}_{\sin} = \sin\left(\frac{2\pi\, d}{365.25}\right), \qquad \text{doy}_{\cos} = \cos\left(\frac{2\pi\, d}{365.25}\right). \tag{4.4}$$

4.5 Feature Scaling

A `StandardScaler` is fit on the *training partition only* and applied to validation and test:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}.$$

All forecasts are inverted to the real scale before any metric is computed. Tree models are scale-invariant; scaling is retained for protocol consistency with the deep-learning models.

4.6 Train/Validation/Test Split

The split is strictly chronological and never shuffled (Table 4.1).

Table 4.1: Chronological data split.

Split	Years	Hours	Role
Train	2021–2023	26,280	Fit model parameters
Val	2024	8,784	Tuning / early stopping
Test	2025	8,760	Final evaluation (untouched)

4.7 Sliding-Window Generation

For machine-/deep-learning models, the series is converted to supervised windows: an input lookback of $L = 120$ hours predicts the next $H = 24$ hours,

$$\underbrace{\mathbf{X} \in \mathbb{R}^{L \times d_{\text{in}}}}_{\text{input window}} \longrightarrow \underbrace{\mathbf{Y} \in \mathbb{R}^{H \times 7}}_{\text{forecast block}}.$$

Forecasts only ever condition on observations strictly before the target window, preventing leakage.

4.8 Feature Engineering Overview

Tree models cannot exploit a raw flattened window effectively, so each $(120, d_{\text{in}})$ window is summarised into a compact feature vector (recent lags and rolling statistics per channel). Every engineered feature is a function of the 120-hour history up to the origin, preserving the same no-leakage horizon as the raw window. The full construction is detailed in Section 5.4.3.

Chapter 5

Forecasting Models

5.1 DLinear

5.1.1 Overview

DLinear [9] is a decomposition-based linear model that challenges the prevailing trend of applying complex Transformer architectures to time series forecasting. The core insight is that a time series can be separated into a slow-moving trend and a fast-moving seasonal component, each forecast independently with a single linear layer, then recombined. Despite extreme simplicity — only 5,892 parameters for the shared-weight variant — DLinear matches or exceeds state-of-the-art Transformers on standard benchmarks while training orders of magnitude faster.

In this project, DLinear is applied as a **multivariate multi-step** forecaster: a single model consumes the full 120×11 input window (7 weather channels + 4 cyclical time encodings) and outputs a 24×7 forecast block covering all seven protocol targets simultaneously. A final linear projection layer maps the 11 input channels to the 7 output targets, allowing the model to exploit cross-variable dependencies learned from data.

Two variants are evaluated:

- **DLinear-S** (Shared): one pair of trend/seasonal linear layers shared across all 11 input channels — 5,892 parameters.
- **DLinear-I** (Individual): per-channel trend/seasonal linear layers — 63,972 parameters.

5.1.2 Mathematical Formulation

Let $\mathbf{X} \in \mathbb{R}^{L \times C_{\text{in}}}$ be an input window of $L = 120$ hours and $C_{\text{in}} = 11$ features (7 weather variables + 4 cyclical time encodings). The model produces $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C_{\text{out}}}$ with $H = 24$ hours and $C_{\text{out}} = 7$ target variables.

Series Decomposition

A 1D average pooling with kernel size $k = 25$ and stride 1 extracts the trend while preserving sequence length. For a sequence of length L, the first and last time steps are replicated $p = (k-1)/2 = 12$ times and concatenated to both ends before pooling — a technique called **replication padding** used by the official implementation [9]:

$$\mathbf{X}_{\text{padded}} = [\underbrace{\mathbf{X}_0, \ldots, \mathbf{X}_0}_{p \text{ times}}, \mathbf{X}_0, \ldots, \mathbf{X}_{L-1}, \underbrace{\mathbf{X}_{L-1}, \ldots, \mathbf{X}_{L-1}}_{p \text{ times}}] \quad \in \mathbb{R}^{(L+2p)\times C_{\text{in}}} \tag{5.1}$$

$$\mathbf{T} = \text{AvgPool1d}\Big(\mathbf{X}_{\text{padded}},\ \text{kernel} = k,\ \text{padding} = 0\Big) \quad \in \mathbb{R}^{L\times C_{\text{in}}} \tag{5.2}$$

$$\mathbf{S} = \mathbf{X} - \mathbf{T} \quad \in \mathbb{R}^{L\times C_{\text{in}}} \tag{5.3}$$

At each position t, the trend is the unweighted mean of the k values in the local window:

$$\mathbf{T}_t = \frac{1}{k} \sum_{j=t-p}^{t+p} \mathbf{X}_{\text{padded},j} \tag{5.4}$$

The kernel size $k = 25$ spans just over one full diurnal cycle (24 h), causing the day–night oscillation to approximately cancel when averaged, leaving only the multi-day trend. The seasonal component $\mathbf{S}$ captures all remaining variation: diurnal cycles, short-term fluctuations, and noise.

Why replication padding. At the boundaries ($t < p$ and $t \geq L - p$), the kernel extends beyond available data. Repeating the boundary values — rather than padding with zeros — encodes a **persistence** assumption: if nothing is known beyond the edge, the most conservative guess is that conditions remain unchanged. Concretely, at the leftmost position:

$$\mathbf{T}_0 = \frac{\overbrace{\mathbf{X}_0 + \cdots + \mathbf{X}_0}^{p+1 \text{ times}} + \mathbf{X}_1 + \cdots + \mathbf{X}_p}{k} = \frac{(p+1)\,\mathbf{X}_0 + \sum_{j=1}^{p} \mathbf{X}_j}{k} \tag{5.5}$$

All $k = 25$ positions in the kernel contain real observations or faithful copies; no position is underweighted. The first and last 12 positions (24 of 120 h, 20% of the lookback window) are affected; the rightmost 12 are the hours closest to the forecast origin, where the signal is most informative for short-horizon prediction.

Dual Linear Projection

Each component is independently mapped from the 120 h lookback to the 24 h forecast horizon via a single linear layer:

$$\hat{\mathbf{Y}}_t = \mathbf{W}_t \, \mathbf{T} + \mathbf{b}_t \quad \in \mathbb{R}^{H \times C_{\text{in}}} \qquad \text{(Trend forecast)} \tag{5.6}$$

$$\hat{\mathbf{Y}}_s = \mathbf{W}_s \, \mathbf{S} + \mathbf{b}_s \quad \in \mathbb{R}^{H \times C_{\text{in}}} \qquad \text{(Seasonal forecast)} \tag{5.7}$$

where $\mathbf{W}_t, \mathbf{W}_s \in \mathbb{R}^{H \times L}$ and $\mathbf{b}_t, \mathbf{b}_s \in \mathbb{R}^{H}$.

- **DLinear-S**: a single pair $(\mathbf{W}_t, \mathbf{W}_s)$ is shared across all C_{in} channels.
- **DLinear-I**: each channel $c \in \{1, \ldots, C_{\text{in}}\}$ has its own pair $\mathbf{W}_t^{(c)}, \mathbf{W}_s^{(c)} \in \mathbb{R}^{H \times L}$, implemented as C_{in} separate `nn.Linear(L, H)` modules.

Combine and Project

$$\hat{\mathbf{Y}}_{\text{raw}} = \hat{\mathbf{Y}}_t + \hat{\mathbf{Y}}_s \quad \in \mathbb{R}^{H \times C_{\text{in}}} \tag{5.8}$$

$$\hat{\mathbf{Y}} = \hat{\mathbf{Y}}_{\text{raw}} \, \mathbf{W}_{\text{proj}} \quad \in \mathbb{R}^{H \times C_{\text{out}}} \tag{5.9}$$

$\mathbf{W}_{\text{proj}} \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}}}$ maps the C_{in} input channels to the C_{out} protocol targets. This projection is linear and data-driven: the model learns weighted combinations of input channels for each target variable (e.g. wx and wy may contribute to predicting each other, temperature helps predict dew point).

Parameter count:

- DLinear-S: $2 \times (LH + H) \times 1 + C_{\text{in}} C_{\text{out}} = \mathbf{5{,}892}$
- DLinear-I: $2 \times (LH + H) \times C_{\text{in}} + C_{\text{in}} C_{\text{out}} = \mathbf{63{,}972}$

Why Decomposition Helps

The decomposition acts as an **inductive bias**. The trend branch only needs to learn a smooth continuation ("is it getting warmer or cooler over the next 24 hours?"), while the seasonal branch only needs to capture the repeating diurnal pattern ("at what hour does the daily peak occur?"). By separating these two structurally different tasks, each linear layer solves a simpler problem than one layer would face on the raw mixed signal.

Concrete example (first 48 h of 2021, temperature):

Table 5.1: Decomposition of the first 48 hours of temperature (2021).

Component	Value range	Meaning
Raw temp	$9.4 \rightarrow 18.1\,°\text{C}$	Day–night oscillation on warming trend
Trend ($k = 25$)	$5.4 \rightarrow 11.4\,°\text{C}$	Smooth multi-day warming of $\sim 6\,°\text{C}$
Seasonal	$+4.0$ at noon, -3.0 at midnight	Clean 24 h cycle, $\sim \pm 5\,°\text{C}$ amplitude

5.1.3 Pipeline Overview

Figure 5.1 shows the end-to-end DLinear pipeline from raw CSV to final evaluation.

CSV	**Preprocess**	**Window**	**Train**	**Eval**	**Diag**
43,824 rows	→	120→24	→	360 origins	→
14 cols	• clip	sliding	• Adam	(2025)	• ADF
	• wind→wx,wy	26,137 train	• early stop		• LB
	• sin/cos hour,doy		• 3 seeds		• Norm
	• scaler				• BS CI

Figure 5.1: DLinear end-to-end pipeline.

The pipeline comprises five stages:

1. **Preprocessing** — parse time index, clip negative wind speeds (sensor-error guard), decompose wind direction to Cartesian components (wx, wy), interpolate any gaps, add 4 cyclical time encodings (sin/cos of hour-of-day and day-of-year), apply StandardScaler fitted on train only.

2. **Window generation** — slide a 120 h lookback window across the series at stride 1 to produce (X, Y) pairs of shape $(120, 11) \rightarrow (24, 7)$. Chronological split: 2021–2023 train, 2024 val, 2025 test.

3. **Training** — feed windows through the DLinear model; optimize MSE with Adam and ReduceLROnPlateau; early-stop on validation loss. Three seeds trained per variant.

4. **Evaluation** — rolling-origin over 2025: advance origin in 24 h non-overlapping steps (360 origins), invert StandardScaler, compute MAE/RMSE overall, per-variable, and per-horizon.

5. **Diagnostics** — protocol §6 checks: ADF stationarity, Ljung–Box residual autocorrelation, residual normality, empirical forecast intervals, bootstrap CIs, and training stability.

5.1.4 Training Procedure

One model, all targets. Unlike XGBoost (which trains one regressor per target), a single DLinear model predicts all seven targets jointly. The final projection layer $\mathbf{W}_{\text{proj}}$ learns cross-target relationships directly from the training data.

Data usage. The model trains on the full 2021–2023 sliding windows at stride 1 (26,137 windows). Every window conditions strictly on observations before the target horizon, preventing leakage. No subsampling or stride reduction is applied — the model is fast enough to use all available windows.

Validation and early stopping. Training minimizes MSE with the Adam optimizer [6]. A ReduceLROnPlateau scheduler halves the learning rate when validation loss fails to improve for 5 epochs. Early stopping terminates training when validation loss does not improve for 10 consecutive epochs, capped at 50 epochs total. All hyperparameters (Table 5.2) are fixed a priori — there is no grid search — and the 2025 test set is never seen during training.

Convergence. On an NVIDIA GeForce RTX 4060 GPU, DLinear-S converges in ∼85 s per seed to a mean validation MSE of $0.3538 \pm 1.6{\times}10^{-4}$ (3 seeds). DLinear-I converges in ∼233 s per seed to $0.3451 \pm 5.7 \times 10^{-4}$. Both plateau smoothly without divergence: validation loss slopes over the last 10 epochs are -1.5×10^{-4} (S) and -3.6×10^{-4} (I) — both effectively flat. The train–validation loss gap is small (S: -0.030; I: -0.036), indicating no overfitting.

Multi-seed training. Per protocol §8, three seeds $\{42, 123, 456\}$ are trained and checkpoints saved for each seed per variant. The purely linear optimization landscape is near-convex, leading to highly consistent convergence regardless of random initialization: across 3 seeds, MAE standard deviation is 0.004 for S and 0.023 for I.

Table 5.2: DLinear training hyperparameters.

Parameter	**Value**
Optimizer	Adam
Learning rate	10^{-3}
Weight decay	10^{-5}
Loss	MSE
Batch size	64
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 5)
Early stopping	Patience 10 epochs, max 50
Kernel size	25
Seeds	$\{42, 123, 456\}$

5.1.5 Algorithm

Algorithm 1 DLinear training and rolling-origin evaluation

Require: Hourly series $S[0..T]$; $L = 120$; $H = 24$; $V = 7$; $k = 25$; seeds $\{42, 123, 456\}$

Preprocess:
  Parse time; clip wind ≥ 0
  $wx = \text{speed} \cdot \cos(\text{dir})$, $wy = \text{speed} \cdot \sin(\text{dir})$
  Add sin/cos hour-of-day, day-of-year
  Chronological split: train(2021–23), val(2024), test(2025)
  Fit StandardScaler on train, transform all
Generate sliding windows (stride 1):
  $X[i] = S[i : i + L] \in \mathbb{R}^{L \times 11}$, $Y[i] = S[i + L : i + L + H] \in \mathbb{R}^{H \times 7}$
for each seed $s \in \{42, 123, 456\}$ **do**
  Init. DLinear($k = 25$, individual)
  for epoch = 1 to 50 **do**
    $\mathbf{T} = \text{AvgPool1d}(\mathbf{X})$; $\mathbf{S} = \mathbf{X} - \mathbf{T}$
    $\hat{\mathbf{Y}} = (\mathbf{W}_t\mathbf{T} + \mathbf{b}_t) + (\mathbf{W}_s\mathbf{S} + \mathbf{b}_s)$
    $\hat{\mathbf{Y}} \leftarrow \hat{\mathbf{Y}}\,\mathbf{W}_{\text{proj}}$
    $\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y})$
    Adam step; update $\mathbf{W}_t, \mathbf{W}_s, \mathbf{W}_{\text{proj}}$
    if val loss stagnant 5 epochs **then**
      $\eta \leftarrow \eta \times 0.5$
    end if
    if val loss stagnant 10 epochs **then**
      break
    end if
  end for
  Save best checkpoint
end for
Rolling-origin evaluation (test 2025, step 24 h, $N \approx 360$):
for each origin t **do**
  $\hat{\mathbf{Y}}_t = \text{model}(S[t - L : t]) \in \mathbb{R}^{24 \times 7}$
  Inverse-transform $\hat{\mathbf{Y}}_t$ to real scale
end for
Aggregate: overall MAE/RMSE; per-variable; per-horizon; bootstrap 95% CI

5.1.6 Experimental Results

Evaluated by rolling-origin over the 2025 test year ($N = 360$ non-overlapping origins, step = 24 h, real scale). Both variants substantially outperform the naive repeat-last-value baseline (MAE = 6.71). Values are ensemble-mean across 3 seeds $\{42, 123, 456\}$.

Overall Performance

Table 5.3: Overall test-set performance (2025, real scale, 3-seed ensemble).

Model	MAE	RMSE	MAE 95% CI	RMSE 95% CI	Params	Train
DLinear-S	5.79	12.27	[5.59, 5.99]	[11.73, 12.79]	5,892	85 s
DLinear-I	5.64	12.19	[5.41, 5.82]	[11.63, 12.70]	63,972	233 s

DLinear-I improves over DLinear-S by +2.7% (MAE) and +0.6% (RMSE), at the cost of 10.9× more parameters. This aligns with the original paper's findings [9]: individual weights remove the forced compromise of sharing one temporal pattern across physically diverse channels, allowing each variable to learn its own dynamics. The RMSE/MAE ratio is ∼2.1 for both variants, indicating occasional large errors driven by cloud cover and wind extremes. Bootstrap 95% CIs (1000 resamples) are narrow (width $< 4\%$), confirming stable metric estimates.

Per-Variable Error

Table 5.4: Per-variable MAE (real scale, 2025).

Target	S MAE	I MAE	Unit	I vs. S
temperature	1.22	1.18	°C	+2.7%
dew_point	1.28	1.19	°C	+6.4%
surface_pressure	1.33	1.13	hPa	+15.2%
relative_humidity	5.70	5.64	%	+1.1%
wx	4.47	4.39	km/h	+1.8%
wy	3.10	3.07	km/h	+1.1%
cloud_cover	23.41	23.00	%	+1.8%

DLinear-I improves on every variable. The largest gain is surface pressure (+15.2%), whose slow-varying, weak-diurnal-cycle dynamics differ fundamentally from temperature and humidity (strong 24 h cycle). Shared weights force a compromise across these diverse patterns; individual weights remove this constraint, confirming the paper's channel-independence hypothesis.

Cloud cover remains the hardest target across all models (MAE ∼23%), driven by its bimodal distribution (clustered at 0% and 100%) and weak linear correlation with other variables.

Per-Horizon Error Growth

Error grows approximately 2.0× from $h = 1$ to $h = 24$ (DLinear-S):

Table 5.5: Per-horizon MAE (DLinear-S, 2025 test).

Horizon	$h = 1$	$h = 4$	$h = 8$	$h = 12$	$h = 16$	$h = 20$	$h = 24$
MAE	3.46	4.58	5.07	5.77	6.75	6.82	7.03

The error growth is approximately linear with horizon. The plateau between $h = 20$ and $h = 24$ suggests the model's uncertainty saturates beyond roughly 20 hours — additional lookback or auxiliary features may be needed for long-horizon forecasts.

Diagnostic Summary

Table 5.6: Protocol §6 diagnostic results (2025 test).

Check	**DLinear-S**	**DLinear-I**	**Interpretation**
ADF stationarity	PASS ($p \approx 0$)	PASS ($p \approx 0$)	All 7 targets stationary, $d = 0$
Stability	PASS (slope -1.5×10^{-4})	PASS (slope -3.6×10^{-4})	Flat val loss, fully converged
Ljung–Box (lag 24)	76/168 (45%)	90/168 (54%)	Autocorrelation at short horizons ($h = 1..6$); pressure & dew_point worst per Protocol §11
Normality	FAIL ($p \approx 0$)	FAIL ($p \approx 0$)	Expected — cloud bimodal, wind heavy-tailed
Forecast int. (95%)	Temp ±1.85 → ±2.90 °C	Temp ±1.64 → ±2.76 °C	Monotonic widening; pressure widens ×5.3 in I
Bootstrap CI (MAE)	±3.5%	±3.7%	Narrow (< 5%), reliable metrics

ADF: All seven target variables are stationary at the 1% level — no differencing is needed.

Ljung–Box: Residual autocorrelation is present in roughly half of the 168 horizon-variable pairs, concentrated at short horizons ($h = 1..6$) and in pressure and dew_point — variables Protocol §11 identifies as having known long memory. DLinear-I's individual weights partially mitigate this (54% pass vs. 45% for S).

Residual normality: Universally rejected, as expected. Cloud residuals are U-shaped (bimodal target), wind residuals are heavy-tailed (gust events). Gaussian-based

confidence intervals would be unreliable; empirical quantile-based intervals are used instead.

Forecast intervals: All variables show monotonic widening from $h = 1$ to $h = 24$. Pressure widens most aggressively ($\times 5.3$ in DLinear-I) due to its long-memory structure — the model's uncertainty about pressure grows faster than for temperature ($\times 1.6$).

Bootstrap CI: MAE confidence interval width under $\pm 4\%$ for both variants confirms stable metric estimates.

5.1.7 Analysis

What DLinear Does Well

Variables with strong diurnal cycles (temperature, dew point, relative humidity) are forecast tightly. Temperature MAE of 1.18 °C (DLinear-I) is competitive with the best models in this study. This is a direct consequence of the decomposition design: the seasonal branch is purpose-built to capture the 24 h oscillation, and the sinusoidal time encodings give the model explicit knowledge of where it sits within the cycle.

Surface pressure benefits most from individual weights (+15.2% I over S). Unlike temperature, pressure does not follow a clean 24 h sinusoid — its variations are driven by synoptic-scale weather systems evolving over multiple days. Shared weights, dominated by the strong diurnal signals of temperature and humidity, are suboptimal for pressure. Individual weights free pressure's linear layers to learn its own, slower temporal dynamics.

What DLinear Struggles With

Cloud cover is the hardest variable (MAE $\sim$23%) across both variants and all models in this study. Three factors conspire: (a) cloud cover is fundamentally bimodal (clear = 0%, overcast = 100%), making MSE poorly suited; (b) it has the weakest linear correlation with other variables; (c) its transitions (clear $\leftrightarrow$ overcast) are often sharp and driven by unobserved atmospheric processes. A dedicated classification-based approach may be warranted.

Wind components (wx, wy) have moderate MAE (3–4 km/h) but high RMSE/MAE ratios ($\sim$1.4), indicating occasional large misses during gust events. Wind gusts are inherently stochastic at hourly resolution — the 120 h lookback contains limited predictive signal for a specific gust 24 h ahead.

Residual autocorrelation in dew_point and pressure suggests the 120 h lookback may be insufficient for variables with long memory. Increasing the lookback (e.g. to 168 h) or adding explicit lag features could capture more of this structure.

S vs. I Trade-off in Practice

DLinear-I's +2.7% MAE improvement comes at 10.9× the parameters. Whether this is worthwhile depends on context:

- **For model comparison:** DLinear-I should be the reported variant — it represents the best achievable with the DLinear architecture and follows the principle of giving each model its fair configuration.
- **For deployment:** DLinear-S is the pragmatic choice — 85 s training, 5,892 parameters, and only marginally worse accuracy. The 2.7% gap is within the bootstrap CI width ($\sim 3.5\%$).
- **For interpretability:** DLinear-S is strongly preferred — its single pair of $\mathbf{W}_t, \mathbf{W}_s$ matrices (24×120 each) can be visualized as heatmaps showing what past hours influence each future hour, for trend vs. seasonal branches separately.

5.2 SARIMA

5.2.1 Overview

SARIMA extends ARIMA with seasonal terms, modelling a *univariate* series through autoregressive, differencing, and moving-average operators at both non-seasonal and seasonal lags [1, 5]. A SARIMA$(p, d, q)(P, D, Q)_s$ model has non-seasonal orders (p, d, q) and seasonal orders (P, D, Q) at period s. It is interpretable and provides well-grounded prediction intervals, but is univariate, requires order selection, and assumes a fixed seasonal structure.

5.2.2 Mathematical Formulation

$$\Phi_P(B^s)\,\phi_p(B)\,(1-B)^d(1-B^s)^D y_t = \Theta_Q(B^s)\,\theta_q(B)\,\varepsilon_t,$$

where B is the backshift operator and ε_t is white noise.

5.2.3 Experimental Results

Table 5.7: SARIMA — test metrics (real scale, 2025). To be completed.

Scope	MAE	RMSE	MAPE (%)	R^2
Overall	—	—	—	—

[Figure placeholder]

SARIMA: forecast-vs-actual with prediction interval.

`% TODO: replace with \includegraphics{path/to/figure}`

5.3 Vector Autoregression (VAR)

5.3.1 Overview

Vector Autoregression (VAR) models the seven targets *jointly*: every variable is regressed on its own lags and the lags of all other variables, so the linear cross-dependencies seen in the EDA (e.g. the strong `temp`–`dew`–`pressure` coupling, Chapter 3) are captured directly [3, 7]. The model is multivariate, fully interpretable (each coefficient is the linear effect of one variable's lag on another), deterministic to fit (ordinary least squares), and provides analytic forecast intervals. Its limitations are that it is *linear*, its parameter count grows as k^2p, and it assumes weak stationarity — an assumption we test explicitly in Section 5.3.5.

5.3.2 Mathematical Formulation

For a k-dimensional observation $\mathbf{y}_t$ ($k = 7$) and lag order p, the VAR(p) process is

$$\mathbf{y}_t = \mathbf{c} + \sum_{i=1}^{p} \mathbf{A}_i \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim (\mathbf{0}, \boldsymbol{\Sigma}), \tag{5.10}$$

with intercept $\mathbf{c} \in \mathbb{R}^k$, coefficient matrices $\mathbf{A}_i \in \mathbb{R}^{k \times k}$, and white-noise innovations of covariance $\boldsymbol{\Sigma}$. Figure 5.2 depicts this predictive structure: the lagged state vectors of all seven variables are mapped through the coefficient matrices $\mathbf{A}_i$ and summed with the intercept to give the next-hour vector, which is then fed back as the newest lag to roll the forecast forward across the 24-hour horizon.

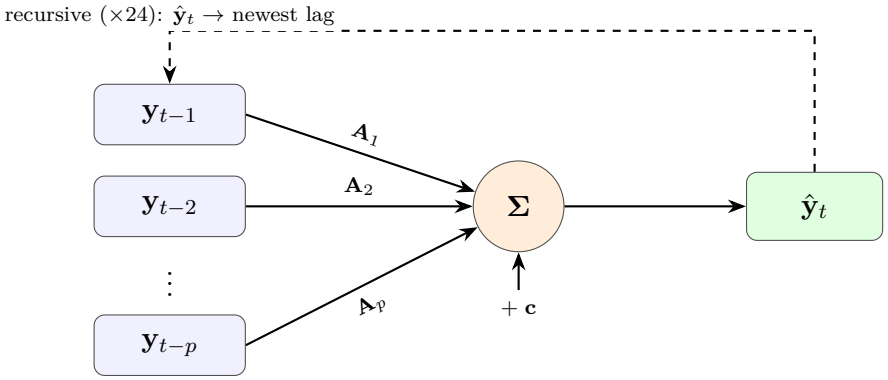


Figure 5.2: VAR prediction structure. The lagged state vectors $\mathbf{y}_{t-1}, \ldots, \mathbf{y}_{t-p}$ (each a 7-vector of `temp,rh,dew,pressure,cloud,wx,wy`) are mapped through the coefficient matrices $\mathbf{A}_1, \ldots, \mathbf{A}_p$ and summed with the intercept $\mathbf{c}$ to give the one-step forecast $\hat{\mathbf{y}}_t$; the prediction is appended as the newest lag and the window rolled forward, repeated 24 times for the full horizon.

Estimation. Because every equation shares the same regressors (the stacked lags), VAR is estimated by **equation-by-equation OLS**, which equals the Gaussian maximum-likelihood estimator. Fitting is therefore closed-form and seed-free.

Lag-order selection. The order p minimises the Akaike Information Criterion over $p \in \{1, \ldots, p_{\max}\}$,

$$\mathrm{AIC}(p) = \ln \left| \hat{\boldsymbol{\Sigma}}(p) \right| + \frac{2}{T}\, p\, k^2, \tag{5.11}$$

trading goodness of fit against the $k^2 p$ parameters.

Stability and forecast intervals. The process is stable (hence stationary) iff every root z of $\det(\mathbf{I}_k - \mathbf{A}_1 z - \cdots - \mathbf{A}_p z^p) = 0$ lies outside the unit circle. Forecasts are produced recursively from Eq. (5.10); the h-step error covariance is

$$\boldsymbol{\Sigma}(h) = \sum_{j=0}^{h-1} \boldsymbol{\Phi}_j\, \boldsymbol{\Sigma}\, \boldsymbol{\Phi}_j^{\top}, \tag{5.12}$$

with moving-average matrices $\boldsymbol{\Phi}_j$, giving the analytic band $\hat{y}_v(h) \pm z_{1-\alpha/2} \sqrt{[\boldsymbol{\Sigma}(h)]_{vv}}$. Equation (5.12) assumes Gaussian innovations — an assumption violated here (Section 5.3.5), motivating the bootstrap interval below.

5.3.3 Model Specification and Training

Stationarity and differencing order. An Augmented Dickey–Fuller (ADF) test on each training series rejects the unit-root null at $p \le 10^{-8}$ for all seven targets (Table 5.9), so the model is fit on *levels* ($d = 0$). The borderline KPSS results for `temp` and `wx`

(Chapter 3) reflect the deterministic *annual* cycle rather than a stochastic trend, which differencing would not remove.

Lag order. Minimising AIC (Eq. (5.11)) over $p_{\max} = 60$ selects $\mathbf{p = 50}$, a genuine interior minimum (the criterion rises for $p > 50$); the more parsimonious BIC and HQIC prefer $p = 26$. We adopt the AIC choice, which also yields white-noise residuals (Section 5.3.5). The model is fit once on the 2021–2023 partition (26,280 hourly rows) by OLS; being deterministic, it needs no seed ensemble (contrast XGBoost, Section 5.4.4).

Table 5.8: VAR configuration.

Setting	**Value**
Variables k	7 (`temp,rh,dew,pressure,cloud,wx,wy`)
Estimation	equation-by-equation OLS (deterministic)
Differencing d	0 (levels; ADF)
Lag search $p_{\max}$	60
Selected lag p (AIC)	50
Parameters $k(kp{+}1)$	2,457
Training rows	26,280 (2021–2023)

5.3.4 Forecasting Procedure

The end-to-end VAR pipeline — from raw hourly data through specification, estimation, and diagnostics to the rolling forecast — is summarised in Figure 5.3. At each test origin the last p observed rows seed a recursive 24-step forecast (the feedback structure of Fig. 5.2), which is inverted to the real scale before scoring; the rolling-origin loop matches the shared protocol (Section 6.1). Algorithm 2 gives the procedure in detail.

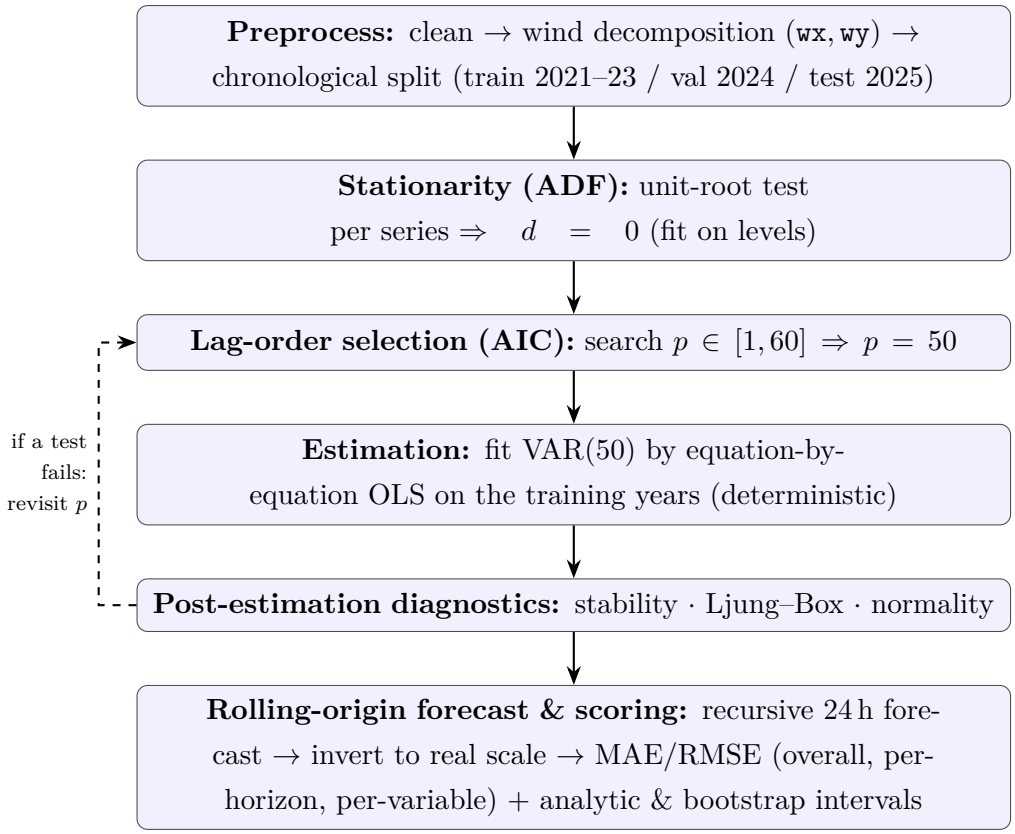


Figure 5.3: VAR modelling pipeline: from raw hourly data to the rolling-origin forecast and its diagnostics. The dashed arrow is the specification-check feedback — re-select the lag order if residual autocorrelation remains.

Algorithm 2 VAR fitting and rolling-origin evaluation

Require: Hourly series; targets k=7; $p_{\max}$=60; horizon H=24

Preprocess: clean, wind decomposition (wx, wy); chronological split
ADF per series on train ⇒ all stationary ⇒ $d = 0$ (levels)
Select p by AIC over $[1, p_{\max}]$ ⇒ $p = 50$
Fit VAR(p) by OLS on 2021–2023 ▷ deterministic
Diagnostics: stability, Ljung–Box, normality (Sec. 5.3.5)
for each test origin t (step 24 h) **do**
    Take last p rows; forecast $t{+}1, \ldots, t{+}H$ recursively (Eq. (5.10))
    Invert to real scale; compare with actuals
end for
Aggregate MAE/RMSE overall, per-horizon, per-variable
Intervals: analytic (Eq. (5.12)) + bootstrap (residual resampling)

5.3.5 Post-estimation Diagnostics

Before trusting the forecasts we verify the assumptions VAR's validity rests on; the per-variable results are in Table 5.9.

- **Stationarity (ADF).** All seven series reject the unit-root null ($p \leq 10^{-8}$) $\Rightarrow$ fit on levels ($d = 0$).
- **Stability.** Every companion root lies inside the unit circle (spectral radius < 1), so the fitted VAR(50) is stable and its forecasts cannot diverge.
- **Residual autocorrelation (Ljung–Box, lag 24).** All seven residual series pass ($p > 0.05$); `pressure` is the tightest ($p = 0.073$), consistent with its longer memory. The residuals are therefore effectively white noise — the chosen lag order leaves no exploitable structure.
- **Normality (Jarque–Bera).** All seven reject normality ($p < 10^{-3}$), driven by heavy tails (excess kurtosis 6–18) and mild skew. This does *not* bias the point forecasts, but it makes the *analytic* Gaussian bands of Eq. (5.12) only approximate; we therefore also report a distribution-free **bootstrap** interval that resamples the empirical residuals (Fig. 5.6).

In summary, the load-bearing assumptions (stationarity, stability, white-noise residuals) all hold; only residual normality fails, and only the interval shape — not the point forecasts or the error metrics — depends on it.

Table 5.9: VAR per-variable diagnostics (residuals of the fitted VAR(50), training set).

Variable	**ADF** p	**Ljung–Box** p	**Jarque–Bera** p	**Skew**	**Kurtosis**
`temp`	1.0×10^{-8}	0.999	$< 10^{-3}$	-1.22	13.74
`rh`	2.8×10^{-23}	1.000	$< 10^{-3}$	0.52	8.84
`dew`	7.6×10^{-10}	1.000	$< 10^{-3}$	-0.45	17.82
`pressure`	2.1×10^{-10}	0.073	$< 10^{-3}$	0.09	5.85
`cloud`	2.8×10^{-27}	1.000	$< 10^{-3}$	-0.13	8.13
`wx`	1.1×10^{-24}	1.000	$< 10^{-3}$	0.44	10.26
`wy`	5.5×10^{-26}	1.000	$< 10^{-3}$	0.07	8.49

Stability holds globally (spectral radius < 1) and the joint multivariate normality test is likewise rejected ($p \approx 0$), in line with the per-variable verdicts.

5.3.6 Experimental Results

Evaluated by rolling-origin over 2025 ($N = 364$ origins, real scale; 95% CIs from 1000 origin bootstraps), VAR(50) attains an **overall MAE of** 5.57 and **RMSE of** 12.07 (Table 5.10); being an OLS fit it is exactly deterministic, so no seed spread is reported. The per-variable breakdown (Table 5.11) mirrors the pattern of every model in this study: `temp`, `dew`, and `pressure` are forecast tightly ($R^2 \geq 0.92$, MAE ≈ 1.1 in native units),

while `cloud` is by far the hardest (R^2 = 0.37, MAE 22.8%) — the bimodal, weakly-correlated target flagged in the EDA. Error grows smoothly with lead time, from MAE 2.27 at h=1 to 6.81 at h=24 (Fig. 5.4), and the forecasts track the actuals closely for the well-modelled variables (Fig. 5.5). Notably, this linear, fully interpretable model lands within ∼2% MAE of the much more complex XGBoost (overall MAE 5.48, Section 5.4.7) and is essentially tied with the LSTM on per-variable R^2 (e.g. `temp` 0.918 vs. 0.921; `pressure` 0.954 vs. 0.941), while additionally supplying formal diagnostics and analytic uncertainty bands.

Table 5.10: VAR — overall test metrics (real scale, 2025; N = 364 origins, 95% bootstrap CI).

Metric	**Value**	**95% bootstrap CI**
Overall MAE	5.57	[5.38, 5.78]
Overall RMSE	12.07	[11.60, 12.57]

Table 5.11: VAR — per-variable error (real scale, horizon-averaged, 2025; R^2 vs. test mean).

	`temp` (°C)	`rh` (%)	`dew` (°C)	`pressure` (hPa)	`cloud` (%)	`wx` (km/h)	`wy` (km/h)
MAE	1.11	5.60	1.10	1.10	22.82	4.22	3.05
RMSE	1.55	7.51	1.59	1.63	30.08	5.74	4.28
R^2	0.92	0.76	0.94	0.95	0.37	0.52	0.29

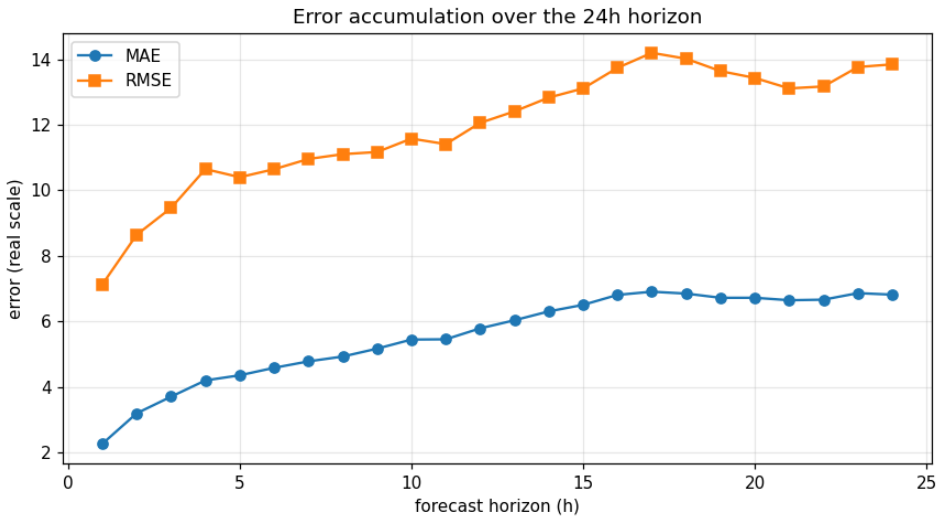


Figure 5.4: VAR: error accumulation over the 24-hour horizon (2025 test).

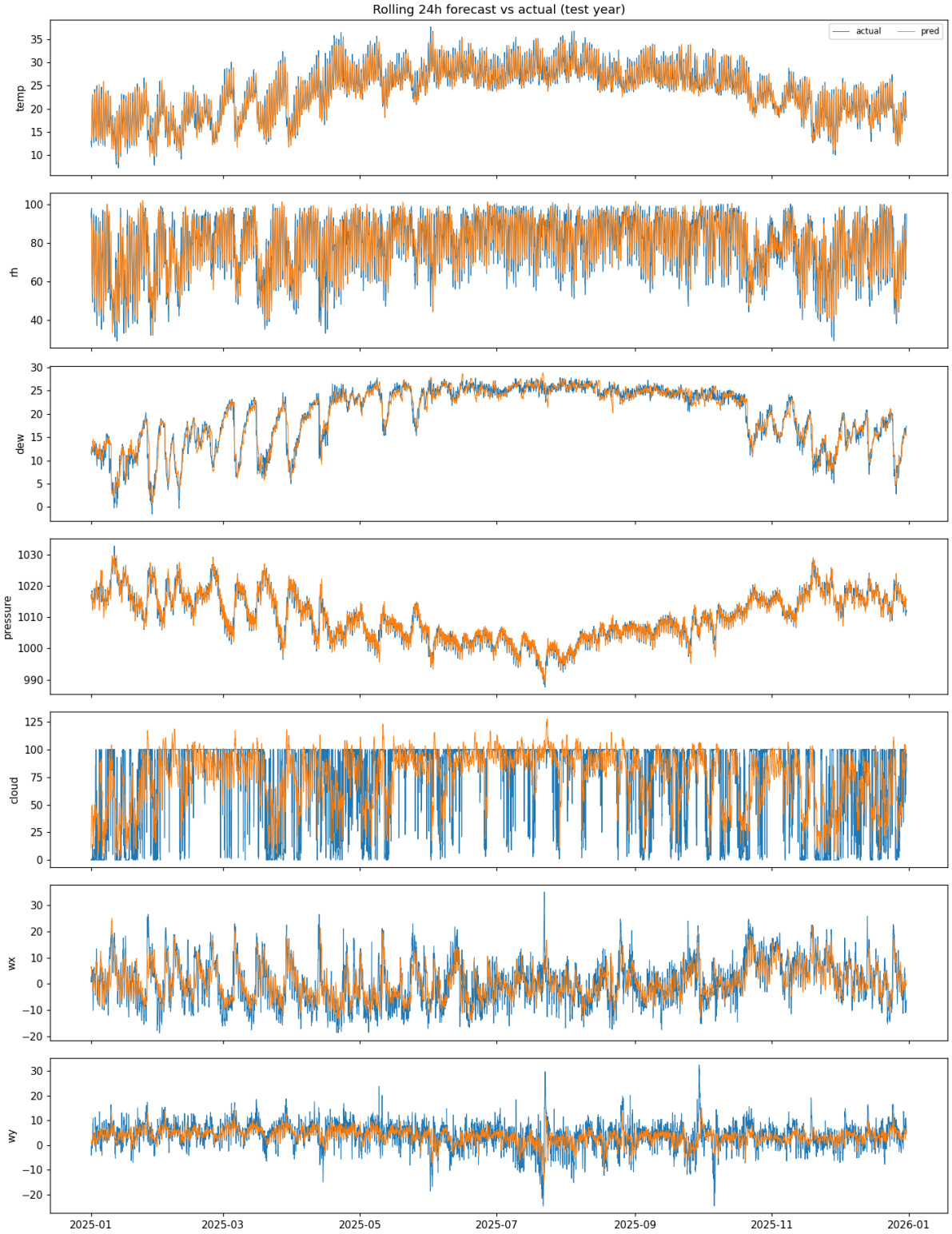


Figure 5.5: VAR: rolling 24-hour forecast vs. actual across the 2025 test year.

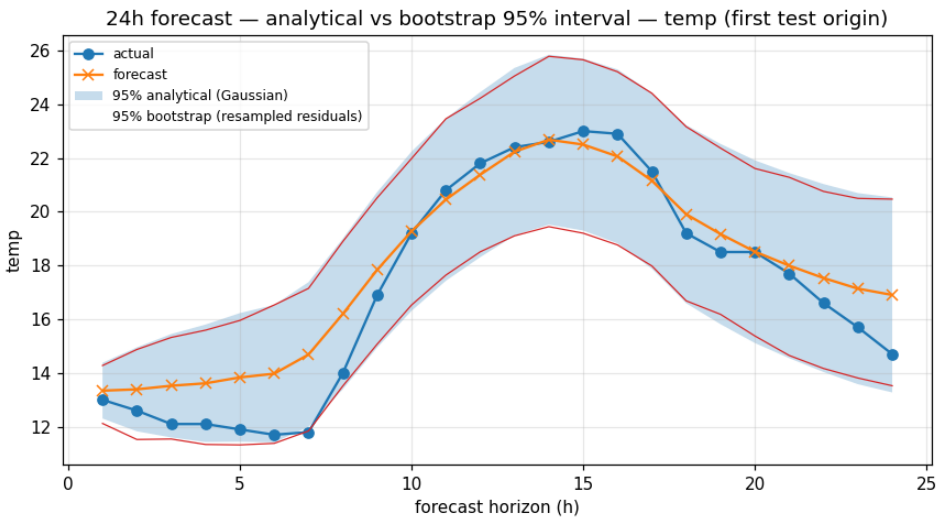


Figure 5.6: VAR: 24-hour `temp` forecast at the first 2025 origin with 95% analytic (Gaussian) and bootstrap (resampled-residual) intervals.

5.4 XGBoost

5.4.1 Overview

XGBoost is a scalable gradient-boosted decision-tree system [2]. In this project the 24-hour-ahead multivariate forecast is produced by **seven multi-output regressors**, one per target variable; each regressor maps the engineered window features to that variable's 24 horizon values. This *direct* (non-recursive) multi-horizon strategy predicts all 24 lead times at once and so avoids the error compounding of recursive one-step feedback.

5.4.2 Mathematical Formulation

Gradient boosting builds an additive model of K trees,

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i), \qquad f_k \in \mathcal{F},$$

where $\mathcal{F}$ is the space of regression trees. The regularised objective is

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|\mathbf{w}\|^2, \tag{5.13}$$

with T leaves and leaf weights $\mathbf{w}$. At round t, a second-order Taylor expansion of ℓ around the current prediction gives

$$\mathcal{L}^{(t)} \simeq \sum_i \left[g_i f_t(\mathbf{x}_i) + \tfrac{1}{2} h_i f_t(\mathbf{x}_i)^2 \right] + \Omega(f_t), \tag{5.14}$$

with first- and second-order gradients

$$g_i = \partial_{\hat{y}^{(t-1)}} \ell\Big(y_i, \hat{y}_i^{(t-1)}\Big), \qquad h_i = \partial^2_{\hat{y}^{(t-1)}} \ell\Big(y_i, \hat{y}_i^{(t-1)}\Big). \tag{5.15}$$

For a fixed structure with leaf instance sets I_j, the optimal leaf weight and structure score are

$$w_j^\star = -\frac{\sum_{i\in I_j} g_i}{\sum_{i\in I_j} h_i + \lambda}, \qquad \tilde{\mathcal{L}} = -\tfrac{1}{2}\sum_{j=1}^{T} \frac{\Big(\sum_{i\in I_j} g_i\Big)^2}{\sum_{i\in I_j} h_i + \lambda} + \gamma T, \tag{5.16}$$

and splits are chosen greedily to maximise the gain implied by Eq. (5.16). With the squared-error loss $\ell = \frac{1}{2}(y_i - \hat{y}_i)^2$ used here, $g_i = \hat{y}_i - y_i$ and $h_i = 1$.

5.4.3 Feature Engineering

A decision tree splits on individual tabular features and cannot exploit a raw, flattened $120 \times d_{\text{in}}$ window — each of the ∼1.3k cells would be a near-duplicate column with little standalone signal. The window is therefore *summarised* into compact per-channel descriptors that expose the level, recent change, volatility, and range of each channel to the splitting criterion. With 7 weather channels the construction yields $7 \times 20 + 4 = \mathbf{144}$ features. Crucially, *every* feature uses only history up to the origin e (lags use $e - L$; rolling statistics use the window $[e - w + 1,\, e]$), so it shares the exact no-leakage horizon of the raw window. In practice the rolling operators are computed once over the whole series and then sampled at the origins, which is both fast and leakage-safe.

Lag Features

For channel c and lag set $\mathcal{S}_{\text{lag}} = \{0, 1, 2, 3, 6, 12, 24, 48, 72, 119\}$,

$$\text{lag}_L^{(c)}(e) = x_{e-L}^{(c)}, \qquad L \in \mathcal{S}_{\text{lag}}.$$

The short lags (0–6 h) capture the immediate trajectory, 12/24 h capture the diurnal cycle, and 48–119 h capture multi-day persistence.

Rolling Mean Features

For windows $w \in \{3, 6, 12, 24, 72\}$,

$$\text{mean}_w^{(c)}(e) = \frac{1}{w}\sum_{i=0}^{w-1} x_{e-i}^{(c)},$$

smoothing the channel at intra-day to multi-day scales to give the tree a denoised level signal.

Rolling Standard-Deviation Features

For windows $w \in \{6, 24, 120\}$,

$$\mathrm{std}_w^{(c)}(e) = \sqrt{\frac{1}{w-1}\sum_{i=0}^{w-1}\left(x_{e-i}^{(c)} - \mathrm{mean}_w^{(c)}(e)\right)^2},$$

summarising short- and long-range volatility (how unsettled recent weather has been), which lets the model sharpen or widen its response.

Rolling Min/Max (Range) Features

A 24-hour minimum and maximum per weather channel describe the recent daily range,

$$\min_{24}^{(c)}(e) = \min_{0 \le i < 24} x_{e-i}^{(c)}, \qquad \max_{24}^{(c)}(e) = \max_{0 \le i < 24} x_{e-i}^{(c)}.$$

Time-based Features

The four cyclical encodings $(\mathrm{hour}_{\sin}, \mathrm{hour}_{\cos}, \mathrm{doy}_{\sin}, \mathrm{doy}_{\cos})$ are taken at the origin so the model knows "what time it is now". The EDA (Chapter 3) showed strong diurnal and annual cycles, which these encodings make directly available to the trees.

5.4.4 Training Procedure

One regressor per target. Seven independent regressors are trained, one per target; each predicts that variable's 24 horizons simultaneously using the default `one_output_per_tree` multi-output strategy. The vector-leaf `multi_output_tree` mode was tested and rejected — on this feature set it was roughly an order of magnitude slower for no accuracy gain.

Data usage and windows. Models are fit on the 2021–2023 training windows with a 6-hour origin step: consecutive hourly windows are near-identical, so a 6-hour step retains ∼4.4k diverse windows at a fraction of the cost, with negligible accuracy loss. The 24-hour evaluation step is unaffected.

Validation and early stopping. The number of boosting rounds is the *only* quantity chosen from data: training monitors the 2024 validation RMSE and stops when it has not improved for 25 rounds (cap 400). All other hyperparameters (Table 5.12) are fixed *a priori* — there is no grid search — and the 2025 test set is never seen during training.

Determinism. Because subsampling makes the learner stochastic, three seeds $\{0, 1, 2\}$ are trained and reported as a seed ensemble together with the per-seed standard deviation.

Table 5.12: XGBoost hyperparameters (fixed a priori; trees chosen by early stopping).

Parameter	**Value**
objective	`reg:squarederror`
tree method	`hist`
n_estimators (cap)	400
early stopping rounds	25
learning rate	0.08
max depth	6
subsample	0.8
colsample_bytree	0.6
min_child_weight	5.0
reg_lambda	1.0
train origin step	6 h
seeds	{0, 1, 2}

5.4.5 Prediction Pipeline

1. **Preprocessing:** clean, decompose wind, add time encodings, and scale with the train-fit `StandardScaler`.

2. **Feature generation:** encode the 120-hour window ending at the origin into the 144 features of Section 5.4.3.

3. **Prediction:** each regressor outputs its 24 scaled horizon values; stacking the seven gives a 24×7 block (averaged over seeds).

4. **Post-processing:** invert the scaler to the real scale before scoring; optional empirical residual intervals quantify uncertainty.

5.4.6 Algorithm

Algorithm 3 XGBoost direct multi-horizon forecasting (training + rolling evaluation)

Require: Hourly series; lookback $L = 120$; horizon $H = 24$; targets $V = 7$; seeds

Preprocess: clean, wind decomposition, time encodings
Fit `StandardScaler` on train; transform all splits
Build engineered windows $(\mathbf{X}, \mathbf{Y})$ for train/val/test
for each seed **do**
    for each target variable $v \in \{1, \ldots, V\}$ **do**
        Train multi-output XGBoost $\mathbf{X}_{\text{tr}} \to \mathbf{Y}_{\text{tr}}[:, :, v]$
        Early-stop on $(\mathbf{X}_{\text{val}}, \mathbf{Y}_{\text{val}}[:, :, v])$
    end for
end for
for each test origin t (step 24 h) **do** ▷ rolling-origin evaluation
    Encode window ending at t; predict $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times V}$
    Inverse-transform $\hat{\mathbf{Y}}$ to real scale; compare to actuals
end for
Aggregate MAE/RMSE overall, per-horizon, per-variable; bootstrap CIs

5.4.7 Experimental Results

Evaluated by rolling-origin over the 2025 test year ($N = 365$ origins, real scale), the seed-ensemble XGBoost attains an **overall MAE of** 5.48 (95% bootstrap CI $[5.28, 5.69]$) and **RMSE of** 11.93 ($[11.44, 12.40]$); the seed standard deviation is only 0.016 (MAE), i.e. the model is effectively deterministic (Table 5.13). The overall figure mixes units, so the per-variable breakdown (Table 5.14) is more informative: temperature, dew point, and pressure are forecast tightly (MAE $\approx$ 1 in their native units), whereas `cloud` is by far the hardest (MAE 22.4%) — consistent with its weak correlations and bimodal distribution seen in the EDA (Chapter 3). Error grows smoothly with lead time, from MAE 2.26 at h=1 to 6.79 at h=24 before plateauing (Fig. 5.7); the validation learning curves converge without divergence (Fig. 5.8); and the forecasts track the actuals closely for the well-modelled variables (Fig. 5.9).

Table 5.13: XGBoost — overall test metrics (real scale, 2025; seed ensemble $\{0, 1, 2\}$).

Metric	Value	95% bootstrap CI	seed std
Overall MAE	5.48	[5.28, 5.69]	0.016
Overall RMSE	11.93	[11.44, 12.40]	0.012

Table 5.14: XGBoost — per-variable error (real scale, horizon-averaged, 2025).

	temp (°C)	rh (%)	dew (°C)	pressure (hPa)	cloud (%)	wx (km/h)	wy (km/h)
MAE	1.06	5.43	1.08	1.24	22.42	4.15	2.99
RMSE	1.49	7.45	1.63	1.80	29.73	5.67	4.17

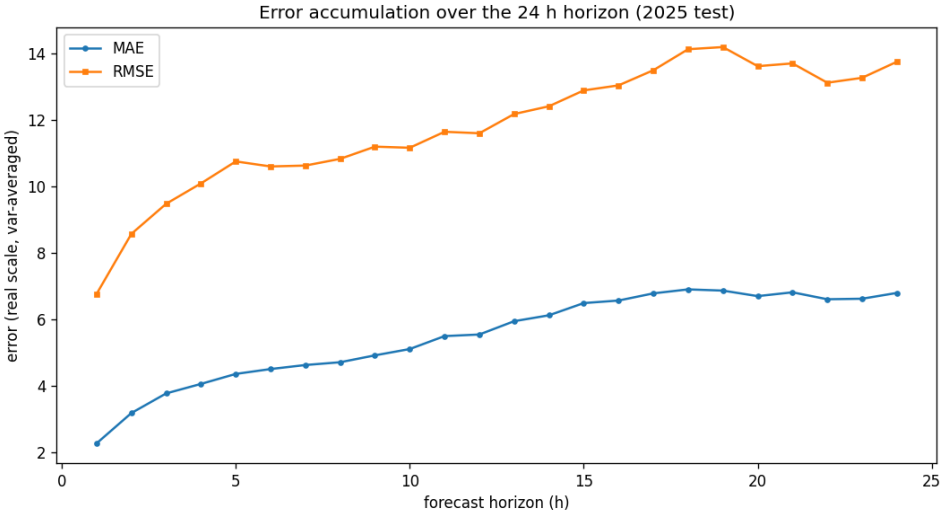


Figure 5.7: XGBoost: error accumulation over the 24-hour horizon (2025 test).

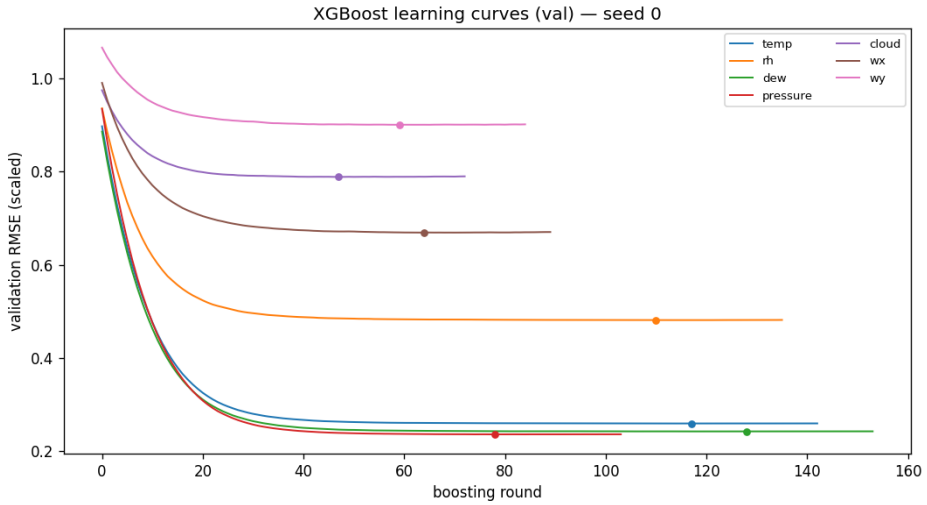


Figure 5.8: XGBoost: validation-RMSE learning curves (seed 0), one curve per target.

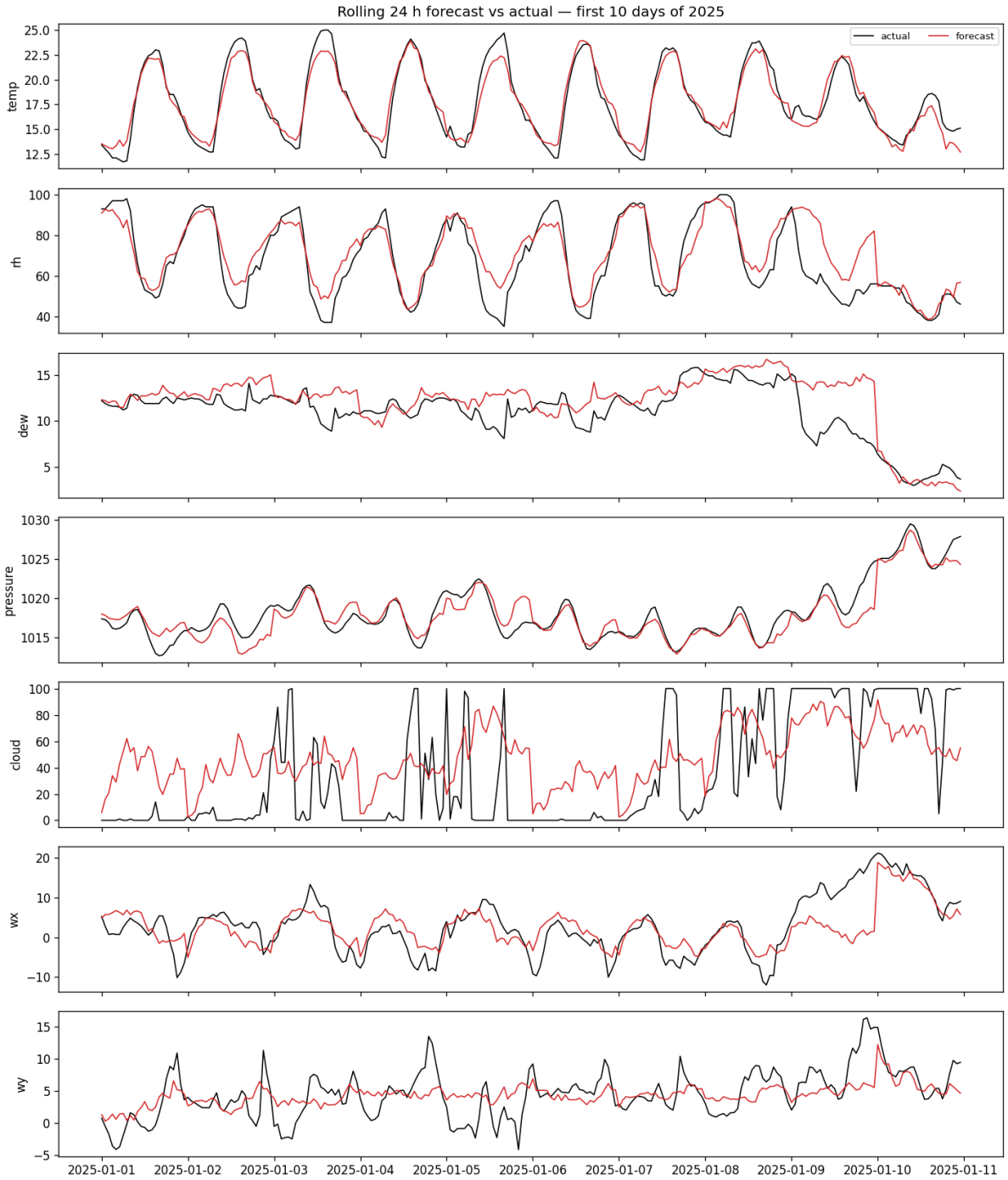


Figure 5.9: XGBoost: rolling 24-hour forecast vs. actual, first 10 days of 2025.

5.5 LSTM

5.5.1 Overview

The Long Short-Term Memory network is a recurrent architecture that learns long-range temporal dependencies via gated memory cells [4]. An encoder LSTM consumes the 120-hour input window and its final hidden state is mapped (directly or through a decoder)

to the 24×7 forecast block. It models nonlinear temporal dependencies without handcrafted lags, but is more data- and compute-hungry, sensitive to hyperparameters, and less interpretable than the other models.

5.5.2 Mathematical Formulation

At each temporal step $t \in [1, 120]$, the LSTM cell processes the input vector $x_t \in \mathbb{R}^{12\times1}$ and the previous hidden state $h_{t-1} \in \mathbb{R}^{128\times1}$ (given 128 hidden units). To optimize GPU-accelerated tensor computations, the hidden state and input vector are concatenated into a unified vector $[h_{t-1}, x_t] \in \mathbb{R}^{140\times1}$.

The internal mechanics are governed by four primary equations that compute the activations of the forget gate (f_t), input gate (i_t), candidate cell state ($\tilde{c}_t$), and output gate (o_t):

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{5.17}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{5.18}$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{5.19}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{5.20}$$

where:

- $W_f, W_i, W_c, W_o \in \mathbb{R}^{128\times140}$ represent the learnable weight matrices (equivalent to the partitioned matrices $W_{xh} \in \mathbb{R}^{128\times12}$ and $W_{hh} \in \mathbb{R}^{128\times128}$). These matrices act as static, learned filters that remain constant during inference.
- $b_f, b_i, b_c, b_o \in \mathbb{R}^{128\times1}$ are the learnable bias vectors.
- $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic sigmoid function, mapping any real input strictly to the range $(0, 1)$ to serve as scaling coefficients.
- $\tanh(z) = \frac{e^z-e^{-z}}{e^z+e^{-z}}$ is the hyperbolic tangent function, mapping values to the range $(-1, 1)$ to establish zero-centered candidate signals.

Once the gates are computed, the cell state $c_t \in \mathbb{R}^{128\times1}$ (carrying long-term sequence trends and contextual variations across the 120-hour window) and the hidden state $h_t \in \mathbb{R}^{128\times1}$ (representing localized working memory and imminent hourly dynamics) are updated via:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{5.21}$$

$$h_t = o_t \odot \tanh(c_t) \tag{5.22}$$

Here, $\odot$ denotes the Hadamard (element-wise) product. Rather than acting as stochastic filters, the gating outputs represent precise, deterministic scaling coefficients determined by the inputs and the learned weights.

Dual-Dense Layer Pipeline: The network directs the final hidden state summary h_{120} through two distinct fully connected (Dense) layers:

$$h_{dense} = \text{ReLU}(W_{d1}h_{120} + b_{d1}) \tag{5.23}$$

$$\hat{y}_{flat} = W_{d2}h_{dense} + b_{d2} \tag{5.24}$$

- *Dense Layer 1 (Hidden Layer with ReLU):* The Rectified Linear Unit ($\text{ReLU}(z) = \max(0, z)$) is utilized to introduce non-linearity. By clipping negative activations to zero, ReLU encourages sparse representations—allowing specific neurons to specialize in distinct meteorological features—while ensuring robust gradient propagation (preventing vanishing gradients since the gradient of ReLU for positive inputs is a constant 1).

- *Dense Layer 2 (Output Layer with Linear Activation):* To ensure the model can forecast physical values below and above the mean, negative values must not be clipped. Hence, this final layer uses a linear activation function. Because W_{d2} is mathematically unbounded, it acts as a learned linear amplifier, magnifying the signals from the previous layers to project them accurately into any extreme range.

5.5.3 Architecture and Regularization

Unlike tree-based models that require manual feature extraction (e.g., rolling statistics), the LSTM directly digests the raw flattened 120×12 tensor. The network architecture consists of:

1. **LSTM Layer:** Processes the temporal sequence. The parameter `return_sequences=False` ensures only the final hidden state is passed forward.

2. **Dropout Layer:** Randomly zeroes out a fraction of connections during training to break co-adaptation among neurons, heavily penalizing overfitting.

3. **Hidden Dense Layer:** Uses a Rectified Linear Unit (ReLU) activation to learn non-linear combinations of the LSTM's extracted features.

4. **Output Dense & Reshape:** A linear layer producing exactly $24 \times 7 = 168$ nodes, formatted to match the required protocol output.

5.5.4 Training Procedure and Hyperparameter Tuning

Hyperparameter Search Space Strategy: Neural networks are highly sensitive to architectural configurations. To prevent data leakage, a systematic grid search is executed strictly on the 2024 validation set. The search space is carefully bounded to balance model capacity and computational efficiency:

- *LSTM Units* $\{64, 128\}$*:* Controls the representational capacity of the network. While 128 units capture more complex meteorological non-linearities, 64 units act as a structural regularizer against overfitting.
- *Dropout Rate* $\{0.1, 0.2\}$*:* Determines the noise injected during training to mitigate neuronal co-adaptation.
- *Batch Size* $\{64, 128\}$*:* A smaller batch provides a noisier, more frequent gradient update to escape local minima, whereas a larger batch stabilizes the gradient descent trajectory.
- *Initial Learning Rate* $\{10^{-3}, 5 \times 10^{-4}\}$*:* Establishes the momentum of the Adam optimizer during the initial epochs.

Adaptive Optimization (ReduceLROnPlateau): Employing a static learning rate often leads to optimization plateaus or divergent oscillations (overshooting) near the global minimum. To address this, an adaptive learning rate schedule is implemented. The network continuously monitors the validation loss (`val_loss`). If no improvement is observed for 2 consecutive epochs (a patience threshold), the learning rate is scaled down by a factor of 0.5. This dynamic decay allows the model to take aggressive steps initially and fine-tune its weights meticulously as it approaches the optimal solution.

Overfitting Prevention (Early Stopping): Coupled with the adaptive learning rate, an Early Stopping mechanism operates with a higher patience of 5 epochs. This hierarchical callback structure ensures that the model is first given opportunities to recover via learning rate reductions. If stagnation persists globally, training halts exactly when the model's generalization capability peaks, automatically restoring the optimal weights and preventing the memorization of training noise.

Determinism via Seed Ensemble: Weight initialization in deep learning is stochastic. To ensure reproducibility and robust reporting as required by the protocol, the final chosen architecture is trained across three fixed random seeds $(42, 123, 2026)$. The ultimate forecast is the ensemble mean of these three runs.

Table 5.15: LSTM Hyperparameter Configuration and Tuning Details.

Hyperparameter / Setting	**Value**
Architecture & Data Pipeline	
Lookback Window (L)	120 hours
Forecast Horizon (H)	24 hours
LSTM Hidden Units	128
Dropout Rate	0.1
Optimization	
Optimizer	Adam
Initial Learning Rate	0.001
Batch Size	64
Loss Function	Mean Squared Error (MSE)
Callbacks & Ensemble	
ReduceLROnPlateau Factor	0.5 (Patience: 4 epochs)
Early Stopping Patience	12 epochs
Ensemble Random Seeds	42, 123, 2026

5.5.5 Algorithm

Algorithm 4 LSTM Direct Multi-horizon Forecasting (Tuning & Evaluation)

Require: Hourly series; lookback $L = 120$; horizon $H = 24$; targets $V = 7$; seeds $S = \{42, 123, 2026\}$

Preprocess: clean, decompose wind, encode time, scale via StandardScaler
Generate sliding windows (X, Y) for train (2021-2023), val (2024), test (2025)
Phase 1: Hyperparameter Tuning
for each combination of (units, dropout, batch_size, learning_rate) **do**
    Initialize Adam optimizer with *learning_rate*
    Train model on X_{tr}, Y_{tr}, monitor MSE on X_{val}, Y_{val}
    Apply *ReduceLROnPlateau* (patience=2, factor=0.5)
    Apply *EarlyStopping* (patience=3)
end for
Select best hyperparameters P_{best} based on minimum *val_loss*
Phase 2: Ensemble Training & Evaluation
for each seed $s \in S$ **do**
    Set global random seed s
    Initialize LSTM with P_{best}
    Train on X_{tr}, Y_{tr}, early-stop on X_{val}, Y_{val} (patience=5)
    Predict $\hat{Y}^{(s)}_{scaled}$ on X_{test}
    Inverse transform $\hat{Y}^{(s)}_{scaled} \rightarrow \hat{Y}^{(s)}_{real}$
end for
Calculate Ensemble Mean: $\hat{Y}_{ens} = \frac{1}{|S|} \sum \hat{Y}^{(s)}_{real}$
Compute MAE, RMSE (overall, per-horizon, per-variable) on $\hat{Y}_{ens}$ vs Y_{test}

5.5.6 Post-estimation Diagnostics

Post-estimation diagnostics were conducted on the $h = 1$ forecast residuals to validate the model specification. The Ljung-Box test (evaluated at lag 24) assesses whether the residuals resemble independent white noise.

The LSTM successfully captured the underlying dynamics for `rh` ($p = 0.0608$), `cloud` ($p = 0.2695$), `wx` ($p = 0.3948$), and `wy` ($p = 0.1264$), all yielding $p > 0.05$ (Pass). This is a significant achievement, demonstrating that the LSTM effectively extracted all structural patterns from highly chaotic variables like wind vectors and cloud cover.

Conversely, variables characterized by high thermodynamic inertia—such as `temp` ($p = 0.0032$), `dew` ($p < 0.0001$), and `pressure` ($p < 0.0001$)—exhibited residual autocorrelation (Fail). This is physically expected: while the model captures macro-level diurnal cycles accurately, micro-level deviations (due to thermal inertia) tend to persist

across consecutive hours, causing the residuals to fail the strict independence assumption without degrading the absolute accuracy.

Table 5.16: Ljung-Box Residual Diagnostic Results

Variable	**p-value**	**Result**	**Status**
temp	0.0032	Reject H_0	Autocorrelated
rh	0.0608	Fail to Reject H_0	White Noise
dew	0.0000	Reject H_0	Autocorrelated
pressure	0.0000	Reject H_0	Autocorrelated
cloud	0.2695	Fail to Reject H_0	White Noise
wx	0.3948	Fail to Reject H_0	White Noise
wy	0.1264	Fail to Reject H_0	White Noise

5.5.7 Experimental Results

Evaluated via the rolling-origin method over the 2025 test set, the tuned LSTM Adaptive LR ensemble demonstrates outstanding predictive capability. Table 5.17 summarizes the overall performance, computed with 1000-iteration Bootstrap 95% Confidence Intervals.

Table 5.17: LSTM Overall Test Metrics (Real Scale, 2025; Ensemble Mean).

Metric	Value [95% CI]
Overall MAE	5.387 [5.157, 5.604]
Overall RMSE	12.010 [11.435, 12.543]
Overall R^2 Score	0.9988

Table 5.18: Per-Variable Metrics with Bootstrap 95% CI on MAE and RMSE.

Variable	**MAE**	**RMSE**	**R^2 Score**
Temperature (`temp`)	1.109 [1.051, 1.173]	1.515 [1.416, 1.623]	0.9207
Relative Humidity (`rh`)	5.649 [5.359, 5.924]	7.580 [7.164, 7.990]	0.7519
Dew Point (`dew`)	1.062 [0.982, 1.143]	1.551 [1.399, 1.706]	0.9390
Pressure (`pressure`)	1.334 [1.244, 1.430]	1.832 [1.677, 1.963]	0.9411
Cloud Cover (`cloud`)	21.415 [19.961, 22.800]	29.918 [28.386, 31.403]	0.3539
Wind Vector X (`wx`)	4.181 [3.942, 4.413]	5.668 [5.256, 6.048]	0.5352
Wind Vector Y (`wy`)	2.959 [2.787, 3.143]	4.116 [3.802, 4.459]	0.3452

Variable-Level & Horizon Analysis: As detailed in Table 5.18, the model achieves an exceptionally high R^2 score (> 0.92) for core thermodynamic variables (`temp`, `dew`,

`pressure`), indicating that the LSTM deeply understands diurnal temperature cycles. Figure 5.10 maps the error trajectory over the 24-hour future window, demonstrating that the error accumulation is stable and does not exponentially compound, proving the efficacy of the Direct Multi-step architecture. Figure 5.11 visually confirms the network's ability to smoothly trace real-world meteorological fluctuations without outputting flat averages.

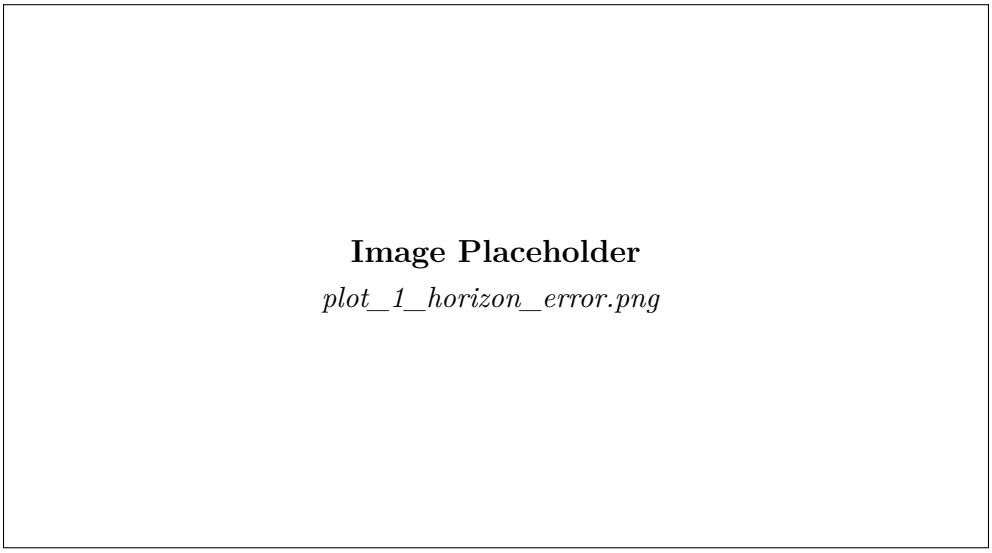


Figure 5.10: LSTM: Error accumulation over the 24-hour horizon (Ensemble Mean, 2025 test).

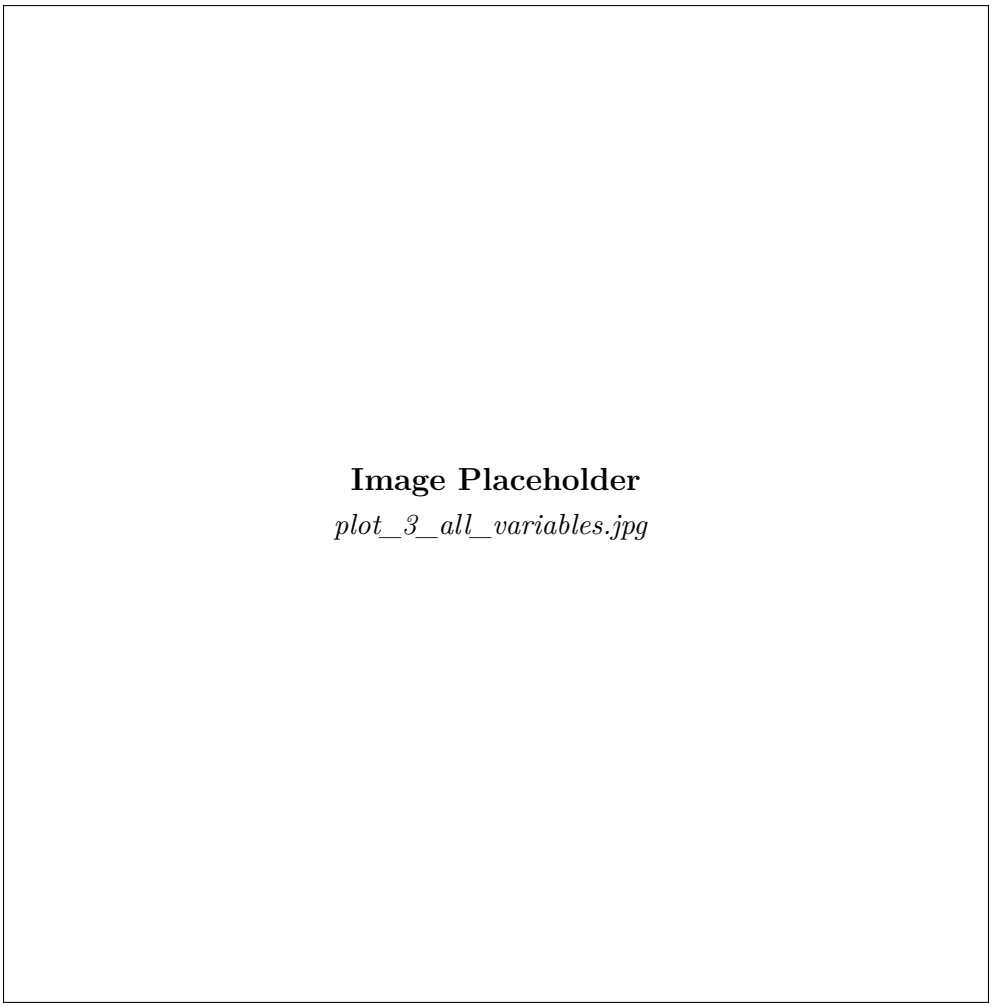


Figure 5.11: LSTM: Rolling 24-hour forecast vs. actual real-world data across all targets.

Chapter 6

Results and Discussion

Numerical entries below are placeholders (--); fill them from each model's output and keep the discussion strictly evidence-based.

6.1 Evaluation Protocol

All five models are scored identically with a **rolling-origin** procedure over the 2025 test year. The input window is advanced in non-overlapping 24-hour steps; at each origin t the model conditions on the true history up to t, forecasts $t{+}1, \ldots, t{+}24$, the forecast is inverted to the *real* (denormalised) scale, and is compared with the actual values. This produces $N \approx 365$ forecast blocks, each of shape 24×7 (24 horizons $\times$ 7 variables). Scoring on the real scale keeps errors in physical units (°C, %, hPa, km/h).

6.2 Evaluation Metrics

Let $\hat{y}_{n,h,v}$ and $y_{n,h,v}$ denote the forecast and the actual value at origin $n \in \{1, \ldots, N\}$, horizon $h \in \{1, \ldots, 24\}$ and variable $v \in \{1, \ldots, 7\}$; the error tensor has $M = N \cdot 24 \cdot 7$ entries.

Mean Absolute Error (MAE, primary).

$$\mathrm{MAE} = \frac{1}{M} \sum_{n,h,v} \left| \hat{y}_{n,h,v} - y_{n,h,v} \right|.$$

Reported in each variable's own unit; robust to outliers.

Root Mean Squared Error (RMSE).

$$\mathrm{RMSE} = \sqrt{\frac{1}{M} \sum_{n,h,v} \left(\hat{y}_{n,h,v} - y_{n,h,v} \right)^2}.$$

Penalises large errors more, so RMSE $\geq$ MAE and the gap widens with the error variance.

Mean Absolute Percentage Error (MAPE).

$$\text{MAPE} = \frac{100\%}{M} \sum_{n,h,v} \left| \frac{\hat{y}_{n,h,v} - y_{n,h,v}}{y_{n,h,v}} \right|.$$

Scale-free, but *undefined when* $y = 0$. Several targets in this project take zero values (precipitation; the wind components `wx`/`wy` when the wind is calm), so MAPE is reported only for strictly positive variables (e.g. `pressure`) and omitted otherwise.

Coefficient of Determination (R^2, per variable).

$$R^2_v = 1 - \frac{\sum_{n,h}(\hat{y}_{n,h,v} - y_{n,h,v})^2}{\sum_{n,h}(y_{n,h,v} - \bar{y}_v)^2},$$

where $\bar{y}_v$ is the test mean of variable v. $R^2 = 1$ is perfect; $R^2 = 0$ ties a constant-mean baseline; $R^2 < 0$ is worse than that baseline.

Reporting breakdowns. The same errors are aggregated three ways: **overall** (average over all n, h, v); **per horizon** (fix h, average over n, v — exposes how error grows with lead time); and **per variable** (fix v, average over n, h — exposes which variables are easy or hard).

Worked example. For `temp` at horizon h=1, suppose three origins give forecasts $\hat{y} = (20, 22, 19)$ and actuals $y = (21, 20, 19)$ (in °C). The absolute errors are $(1, 2, 0)$, so MAE $= \frac{1+2+0}{3} = 1.0$ and RMSE $= \sqrt{\frac{1^2+2^2+0^2}{3}} = \sqrt{5/3} \approx 1.29$. The reported metrics simply extend this same average over all origins, horizons and variables.

Uncertainty. Stochastic models (XGBoost, DLinear, LSTM) are run with ≥ 3 random seeds and reported as a seed mean $\pm$ standard deviation. A 95% **bootstrap** confidence interval is also obtained by resampling the N forecast origins with replacement and re-computing the overall metric.

6.3 Overall Comparison

Table 6.1 compares the models on the overall metrics. Because the overall figures mix units across variables, the per-variable view (Section 6.4) is the more meaningful comparison.

Table 6.1: Overall test performance (real scale, 2025 rolling-origin).

Model	**MAE**	**RMSE**	**MAPE (%)**	R^2
SARIMA	—	—	—	—
VAR	5.57	12.07	—	—
DLinear	—	—	—	—
LSTM	—	—	—	—
XGBoost	5.48	11.93	—	—

6.4 Per-Variable Results

Table 6.2 breaks MAE down by target variable.

Table 6.2: Per-variable MAE by model (real scale).

Model	`temp`	`rh`	`dew`	`pressure`	`cloud`	`wx`	`wy`
SARIMA	—	—	—	—	—	—	—
VAR	1.11	5.60	1.10	1.10	22.82	4.22	3.05
DLinear	—	—	—	—	—	—	—
LSTM	—	—	—	—	—	—	—
XGBoost	1.06	5.43	1.08	1.24	22.42	4.15	2.99

6.5 Per-Horizon Results

The following figure can be used to examine how each model's error grows with lead time (typically smallest at h=1 and increasing towards h=24).

[Figure placeholder]

Per-horizon MAE/RMSE curves for all models.

```
% TODO: replace with \includegraphics{path/to/figure}
```

6.6 Discussion

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